VisiTECH Data Acquisition Package

User's Guide

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1. Introduction to DA Package:

1.1 Welcome to DA Package:

This Data Acquisition Package provides complete solution for data acquisition, presentation, analysis and storage for various process industry applications.

This package is designed ready to use. Most of the common screens are created and they are available for just a click on menu item or a button on toolbar item. This package is developed with high performance language (VC++) and data storage is also done in more optimized way.

All data is stored with its time stamp, so keeping track of time is a snap too.

1.2 System Requirements:

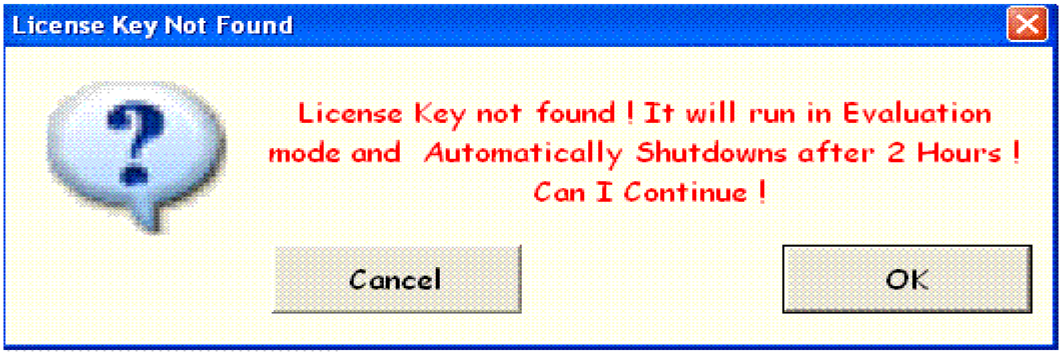
Any x86 based computer capable of running Windows 2000, XP or later versions of OS. We suggest at least P-II 350 M.Hz Processor, 64 MB of RAM and 20 MB of hard disk space. The program is designed to be responsive even under processor strain.

To run this package, monitor resolution should be 1024 X 768. So your system monitor required to be set to this screen resolution before starting the DA Package.

1.3 Licensing DA Package:

This package comes with a hardware key. Without a key if you start the package then it will run in a demo mode for 2 Hours and the end of 2 hours it will automatically shut downs.

If you start the DA package without Hardware key then the following dialog will prompt to take feed back from the user either to switch to evaluation mode or close the package.



To get the Hardware key you can contact your DA package provider.

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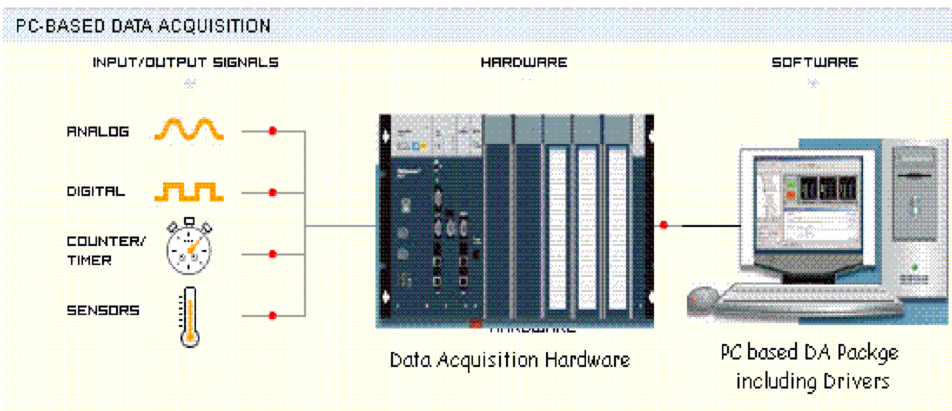
1.4 What is Data Acquisition?

Data acquisition is the process of gathering process da in an automated fashion from analog and digital measurement sources such as sensors and other devices. Data acquisition uses a combination of PC-based measurement hardware and software to provide a flexible, user-defined measurement system.

Basically, any computer based Data Acquisition system consists of Acquiring, Monitoring the process data, Presenting, Analyzing and finally Storage it for further processing. This DA package provides support for all the above functionalities.

The block diagram of PC based Data Acquisition system is shown in below fig:



At presently this DA packages provides the following set of protocols to communicate with DA Hardware (*Including PLC's, PID Controllers, DA Cards, etc.,*). The below set of protocols communicate with Hardware either over Serial bus or Ethernet.

- *Modbus Serial –RTU*
- *Modbus Serial – ASCII*
- *Modbus Ethernet*

In addition to above protocols this package supports OPC Client interfaces. So any of your hardware device have valid OPC DA Server then you can easily communicate with your device through OPC Client.

To develop your custom drivers for your proprietary protocols over your choice of communication medium (*i.e., GPIB, USB, ZigBee, PCI, PXI, etc.,*) we will provide Driver Development wizard. So you can easily build the driver and start acquiring the data with this package.

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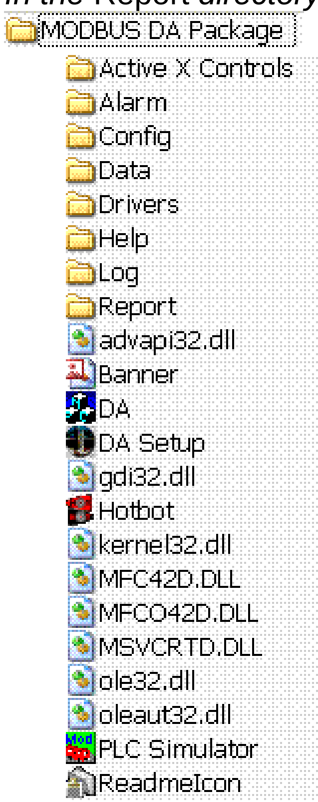
1.5 DA Package Installation and Directory Structure:

Once running the installation software, you can accept most of the default settings. In the Installation Folder dialog you can enter the location for this DA Package.

Once Installation completed, in the installation folder the following directory structure will be created:

- *In the* Alarm *directory Alarm & Event related files are stored.*
- *In the* Config *directory all sets of configuration files will be saved.*
- *In the* Data *directory, Historic data will be stored.*
- *In the* Drivers *directory supporting communication drivers will be reside.*
- *In the* Help *directory, all help files will be reside.*
- *In the* Log *directory every day log report will be stored.*
- *In the* Report *directory the generated reports will be reside.*



In addition to this, short cut in the program folder will be created and a Icon on the desktop will be created.

Take note only one instance of this package will run in a system*. (i.e. you can instantiate multiple instantiations of this package in a single system*).

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1.6 Abbreviations:

The following abbreviations used in this document:

AI : Analog Input
ALM : Alarm
AO : Analog Output
DCS : Distributed Control Systems
DI : Discrete Input
DO : Discrete Output
EVT : Event
GT : Greater Then
HI : High (*Used with Alarm configurations*).
HI HI : High High (*Used with Alarm configurations*).
LT : Less Then
LO : Low (*Used with Alarm configurations*).
LO LO : Low Low (*Used with Alarm configurations*).
MSG : Message
OLE : Object Linking & Embedding
OPC : OLE for Process Control
PLC : Programmable Logic Controller
PV : Process Variable. Normally AI read value.
RO : Read Only
ROC : Rate Of Change.
RW : Read Write
SCADA : Supervisory Control and Data Acquisition

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2. DA Configuration Dialogs:

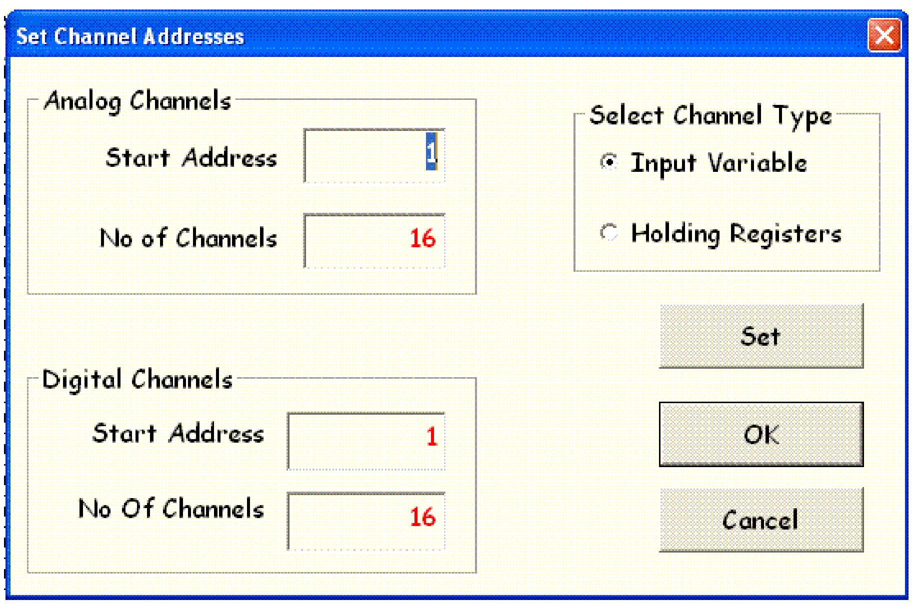
2.1 Channel Address Configuration:

You can access the *Channel Address Configuration* dialog by clicking on *Set Address* button on Front panel.

This Channel Address Configuration dialog is available for Administrative privileged users only. For more on Administrative privileges see the Login Management chapter.

Once communication with device is established this *Set Address* button is disabled. Once disconnected the communication this button is automatically enabled if the currently login user have administrative privileges.

In this Address Configuration dialog you can set the **Start Address** and the **No of Registers**. Address configuration dialog is shown in below fig: You can easily switch between Read Only channel types and Read Write channel types by selecting the different options in the Select Channel Type radio button.



AI: Analog Input or Read only registers and each register will hold 16 bit value to get the data from external source.

DI: Discrete Input or Read only Discrete Inputs. Each register will contain single bit to get the data from external source.

AO: Read Write Output registers or Holding Registers. Each register consists 16 bit numeric number is used to get or set the data from external device.

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DO: Read Write Discrete Outputs or Output Coils Consists 1 bit length used to drive output data to a digital output channel.

Example: *Suppose if you want to select* **AI Addresses** *start from* ***100*** *and No of channels as* ***20****. Just select Channel type as* **Input variable** *and enter address text box* **100** *and in No of channels text box* **20** *and click on* **Set** *button.*

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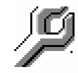
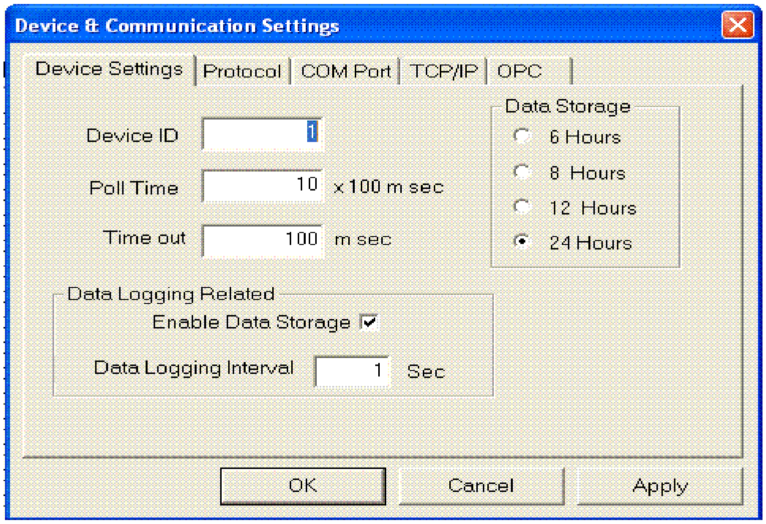
2.2 Device Configuration:

This Device Configuration Property sheet consists of set of pages. The first page consists of Device related configuration parameters. Users can display this property sheet in any one of the following methods:

- *By clicking on the menu item* Configuration → Device & Communication Settings.

- *Clicking the* Device & Communication Settings *tool bar button* 



- **Device ID:** In this Device ID text box you can enter connected device ID (number only).
    - o For MODBUS complaint devices these numbers are between 1 and 247.

- **Poll Time:** Enter the poll time in 100's of milli seconds. Suppose if you entered as 20 then once in 2 seconds it will scan the data from device.

- **Time Out:** Enter the timeout value in terms of milli seconds. This configuration filed used while communicating with device.

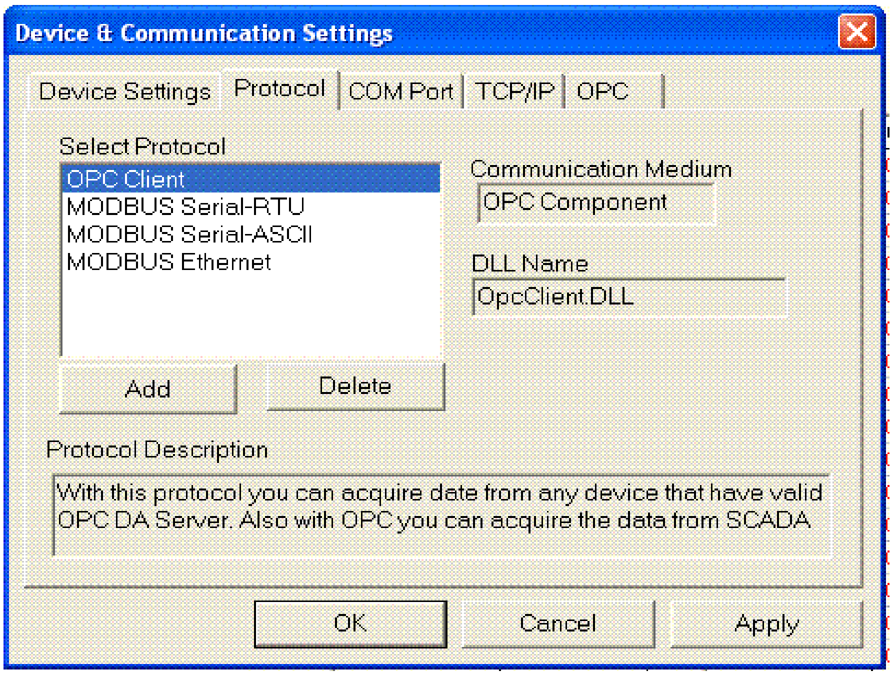
- **Enable Data Storage:** *With this check box you can either enable or disable the data storage.*

- **Data Logging Interval:** The data logging interval will be in terms of seconds only. Support if your Poll time is 500 m.sec and Data Logging interval as 10 seconds then the data will be stored once in every 20 scans. These 20 scans will be used for alarm generation and monitoring purpose.

- **Data Storage:** In this radio buttons you can specify the data storage interval. Once that interval is finished the data will be stored in a file and the further data will be stored in different file.

2.3 Protocol Configuration:

The second property page of the Device Configuration Sheet is the **Protocol Configuration** page. In this property page you can select the communication protocol for the connected device. In addition to this you can add/delete new protocol drivers.

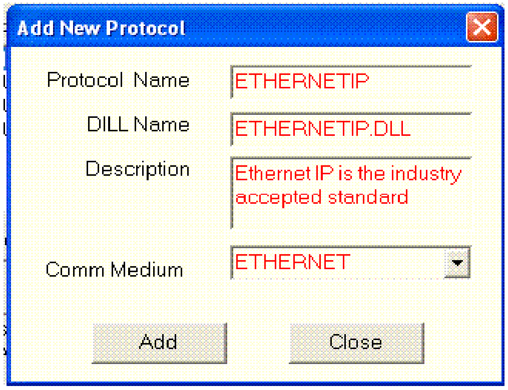


By moving the UP/DOWN arrows you can switch between different protocols. Protocol related information will be displayed in the Communication Medium, DLL Name and Protocol Description text boxes.

You can build protocol drivers for your custom devices. Those drivers you can easily by clicking the add button. To build drivers we will provide driver development wizard. Once you build the driver just copy that DLL to *Drivers* directory within in you *DA Package installation directory* then you have to add it to the existing Protocol driver list.

Users can easily find the Drivers *directory in the Location Dialog. This dialog is activated by clicking on the* Settings→Paths *menu item.*

Once you click on the Add button the following dialog will be displayed. Here you can add your interested name for your communication protocol driver and this name will be displayed in the available protocol list. Then you need to add DLL name. This DLL is required to copied into the *Drivers* directory. Then you need to provide short protocol description and finally you need to select communication medium type. To specify the communication medium type you have the following options SERIAL, ETHERNET, OPC and OTHER.



If your communication driver communicated with devices other then SERIAL, ETHERNET, OPC like USB, GPIB, etc., then you have to select OTHER. If you select OTHER you don't get any kind of configuration dialogs all the communication specific settings will be either hard coded in the driver it self or you have to generate INI files and use them.

At present we are providing the configuration dialogs for SERIAL, ETHERNET and OPC drivers only. In the future we can add the configuration dialogs for USB, PCI, PXI devices also.

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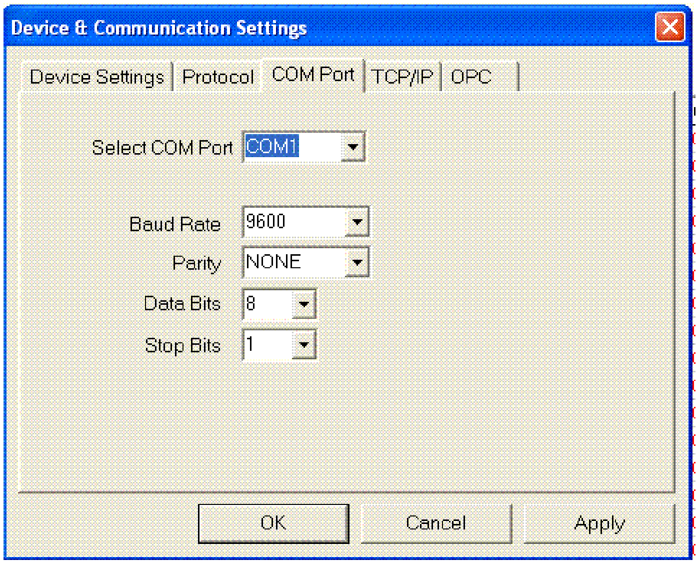
2.4 Serial Port Configuration:

The third property page of the Device Configuration Sheet is the **Serial Port Configuration** page. In this property page you can select the Serial Port. (*At present we are supporting maximum of 4 serial ports* **COM1** *to* **COM4***).* You can set each serial port configuration parameters such as **Baud Rate, Parity, Data bits,** and **Stop bits**. The Serial port configuration dialog is shown in below fig:



Note: For Modbus RTU protocol always set the Data Bits as 8.

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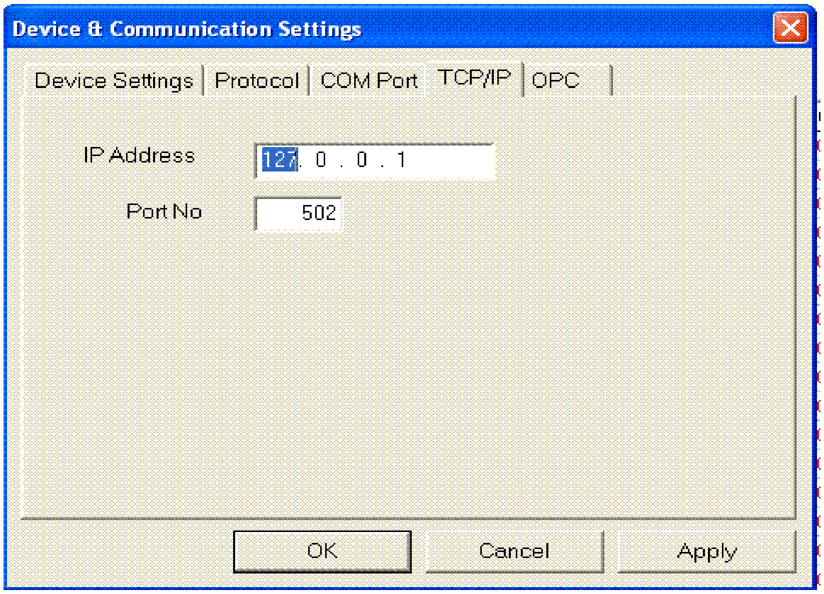
2.5 Ethernet Port Configuration:

The fourth property page of the Device Configuration Sheet is the **Ethernet Port Configuration** page. In this property page you can set the IP Address and IP Port no of your device. The configuration dialog is shown in below fig:



Note: For Modbus complaint devices the default port is 502.

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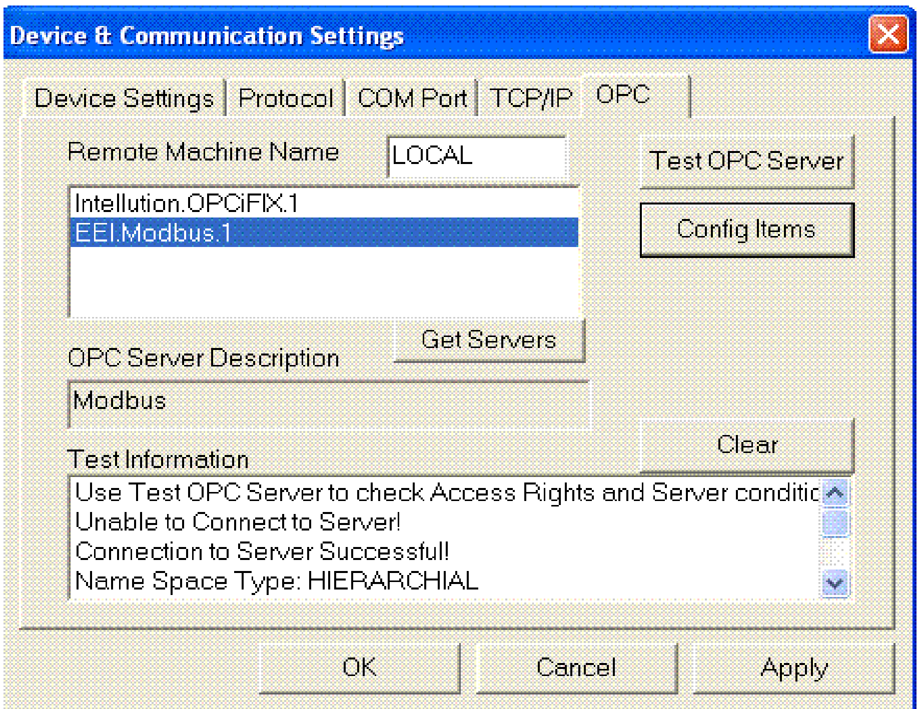
2.6 OPC Server Configuration:

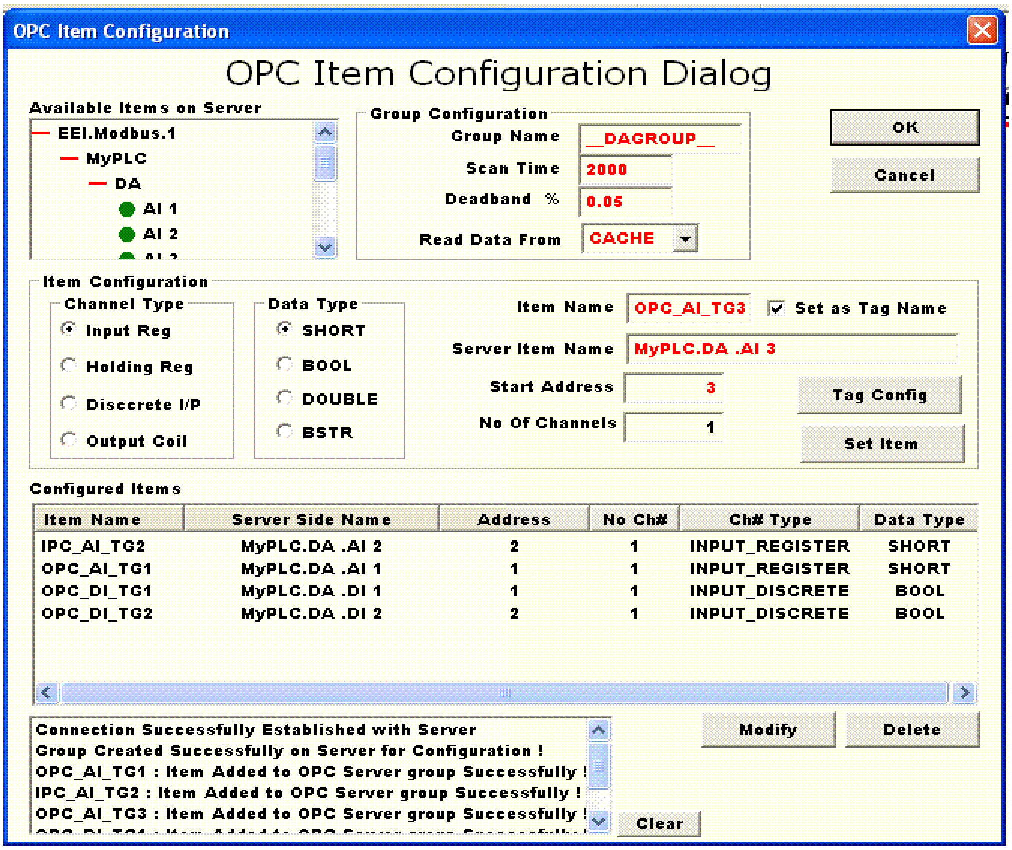
The last property page of the Device Configuration Sheet is the **OPC Configuration** page. In this property page you can select the OPC Server either on local machine or over remote machine. Suppose if your server resides on local machine (*i.e., On the same machine where DA Package is running*). Then you can enter the remote machine name as local or 127.0.0.1. If your OPC Server resides on remote machine then you can enter the remote machine name or its IP address.

Once entered the Remote machine name just click on Get Servers button. It will enumerate all the available **OPC DA Servers** installed on the specified machine. In this servers list you can select Server and test for availability and security related constraints.



2.7 OPC Item Configuration:

In this OPC Item configuration dialog you can configure OPC Items on the selected OPC Server. The OPC item configuration dialog is shown in below fig:



In this configuration dialog you can see all the available Items on selected OPC server will be displayed in *Available Item On server Tree Control*. Here Items are arranged hierarchically the same way they are configured over OPC Server.

Initially you need to specify group name that need to be create at OPC server for your DA Package configured items. You can also specify the **Scan Time, Deadband** and **Read Data From** options also.

You can select the Items by double clicking on any item in the tree control. That Item full name will be displayed in the Server Item name text box. Then you have to specify the channel type (*i.e., AI, DI, AO or DO*) also need to select Data type. Then you need to specify Item name. This name is visible at client side (i.e., you can't see from OPC Server). Basically this name is an alias for Server Item name. Finally you need to specify Start address and No of channels. These address

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value should be within the range you specified in the address configuration dialog.

If you selected the Set as Tag Name check box Then this Item name is set as Tag Name for that address.

Once clicked on Set Item button, the it will first adds this item to OPC Server. If it is successfully added then this item will be added to OPC configuration list. All the configured items you can find in the bottom list box.

Once configured the items you can modify or delete the items.

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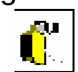
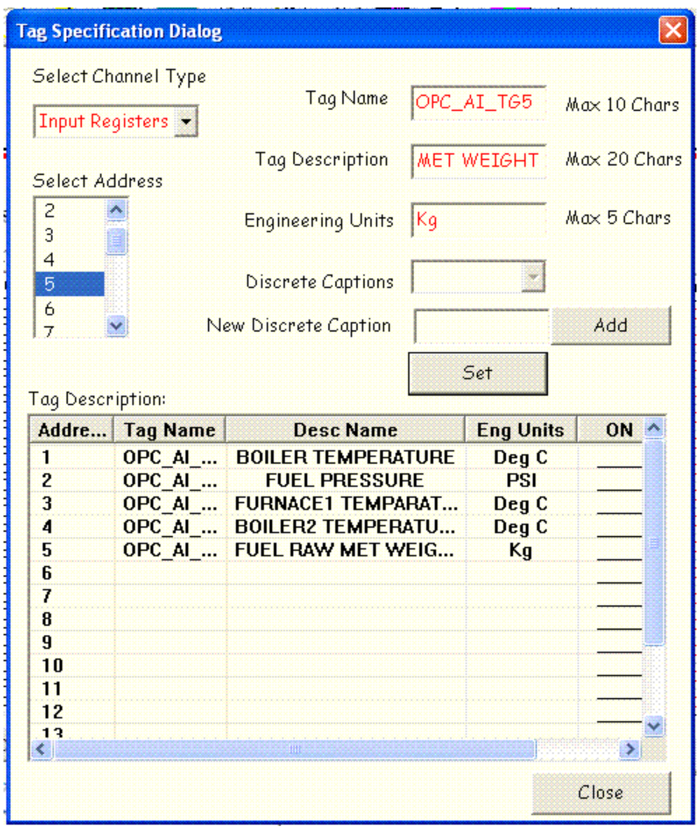
2.8 Tag Configuration:

With this Tag Configuration Dialog you can set the Tag Name, Descriptive Name, Engineering units (*For Analog Channels Only i.e., AI & AO channels*) and ON Caption, OFF Caption (*For Discrete Channel i.e., DI & DO channels*).

Users can display this Tag Configuration dialog in any one of the following methods:

- *By clicking on the menu item* Configuration → Tag Configuration
- *By Clicking the* Tag Configuration *tool bar button* 

The Tag Configuration dialog is shown in below fig:



Channel Type: In this combo box you can find four different kinds of channels Input Registers (AI), Holding Registers (AO), Input Discrete (DI) and Output Coils (DO). Once you selected any one of the above channel types then it will automatically update the Address list box and Tag description lists.

Address: This list box contains the all the channel addresses of the currently selected Channel Type.

Tag Name: You can set maximum of 10 characters Tag Name for each channel in all channel types.

***Example**: Tag Name for any AI Channel BLR1_TMP, Similarly DI channel is OPC_DI_TG1, etc.,*

Descriptive Name: With this text box you can set maximum of 20 characters descriptive name for all kind of channels.

Example: The Descriptive name for any of the channel is "*ETHYLE AMINE BLR TEMP*", "*NITROGEN MIXTURE TEMP*", etc.,

Engineering Units: You can set the engineering units only for Analog channels (*AI & AO i.e., Input Registers & Holding Registers*). This text box will be enabled if you selected one of the Analog Channel in Channel type Combo box.

New Discrete Caption & Discrete Caption: This Discrete Caption will be set for only Discrete Channels (*DI & DO i.e., Discrete Input & Output Coils*). This Combo box and text box will be enabled if you select one of the Discrete Channel Type in the Channel type Combo box.

To add new discrete caption you need to enter the New Discrete Caption text box like *OPEN/CLOSE* format. (*Here OPEN stands for Boolean TRUE or 1, similarly CLOSE is used for Boolean FALSE or 0).* Once you click Add button this new discrete caption will be available in Discrete Caption Combo box. Now it is ready to select for any Discrete Channels.

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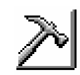
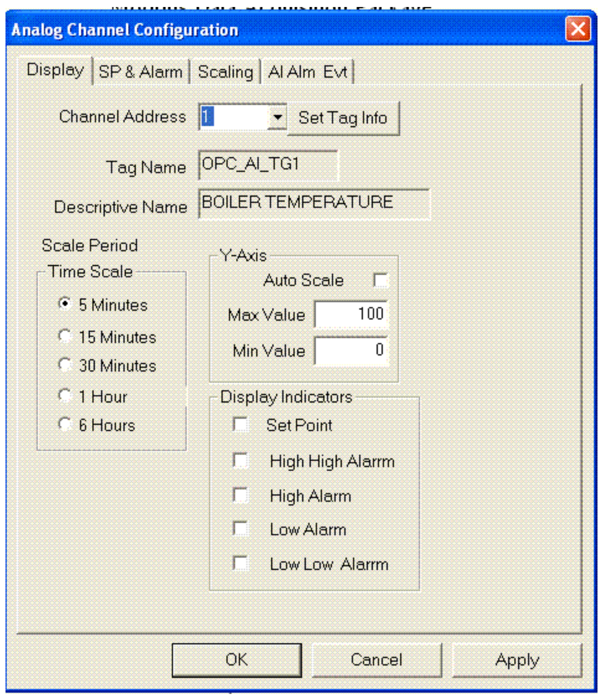
2.9 Analog Channel Configuration:

Analog Channel Property sheet consists of three property pages. All these property pages are used to configure only AI (*Analog Input or Input Register*) channels only. This configuration dialogs are used to set Maximum, Minimum values, you can define various Alarm values and any Signal condition specific settings.

Users can display this Analog Channel Configuration dialog in any one of the following methods:

- *By clicking on the menu item* Configuration → Analog Channel Configuration.
- *By clicking the* Channel Cfg *button on the Front panel.*
- *By Clicking the* Analog Channel Configuration *tool bar button* .

First property page in this Property Sheet is shown in below fig. Once you selected any one of the AI channel then its Tag Name and Descriptive Name automatically displayed these tag descriptions you can see from all three property pages.



Auto Scale: If you check this check box, then Maximum and Minimum text boxes will be automatically disabled. In trend view the Maximum and Minimum values are automatically calculated at runtime from the collected data. If any of the new data comes beyond the Max, Min limits then these Max, Min values are automatically updates.

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Max Value: You can set the Maximum AI value for the currently selected AI Channel. This text box is available if Auto Scale Check box is not checked. Unlike Auto Scale mode this value can't be updated at run time.

Min Value: You can set the Minimum AI value for the currently selected AI Channel. This text box is available if Auto Scale Check box is not checked. Unlike Auto Scale mode this value can't be updated at run time.

Time Scale: This value will be used to specify the each trend screen size. You can set different trend screen times for different AI Channels.

Display Indicators: These check boxes used to select the display indicators in the trend view. These check boxes allow you to select only properly configured alarms only. For alarm configuration you can find second property page of this Analog Channel configuration Property sheet.

The Second Property Page in this Property Sheet is shown in below fig. With this property page you can set Set Point and various Alarm values. In *Auto Scale* mode you can't set any of these Alarm values.

Before setting the Set Point and Alarm Conditions you need to set proper Maximum and Minimum values for corresponding channel. You can set different Set point and Alarm values for each channel.

The Select Configuration parameter Combo box allows you to select any one of the configuration parameter. This SET POINT, HI HI alarm Value, Hi Alarm Value, Lo Alarm Value & Lo Lo Alarm values should be within the Maximum and Minimum values those are set in the first previous property page for the corresponding AI channel.

- SET POINT
- HIGH HIGH ALARM VALUE (Alarm)
- HIGH ALARM VALUE (Warning)
- LOW ALARM VALUE (Warning)
- LOW LOW ALARM VALUE (Alarm)
- RATE OF CHANGE ALARM VALUE (Alarm)

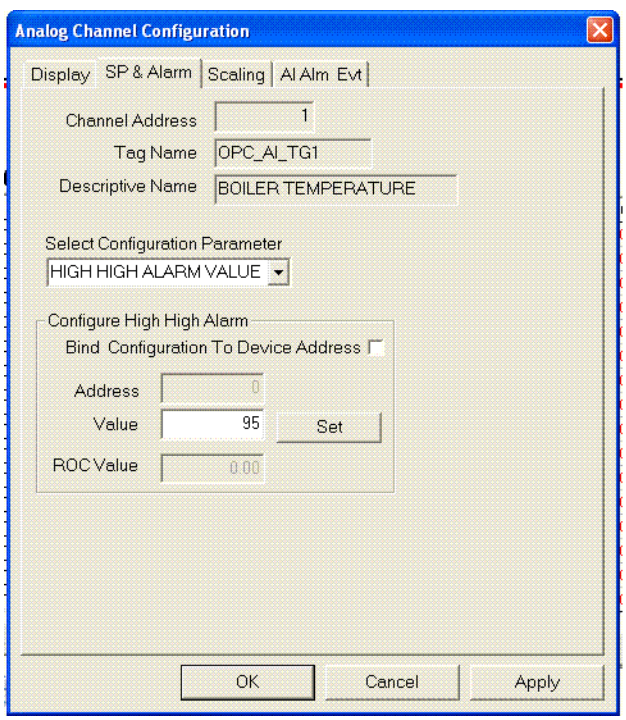
From the above first five configurations you can set in two ways:

- Binding to Device Address: If you select *Bind Configuration to Device Address* check box then the address text box will be enabled. In this text box you need to specify the device AO (or Read write address) where the configuration is defined. Once you entered the address if you click on *Set* button then value will be read from the specified address from the connected device. If the values successfully read then this particular configuration is set

other wise an Error message box is prompted and configuration is not set.
Note: *Before setting the configuration you need to make sure the package is connected to the device and communication is established.*

- Directly Setting the Value: if *Bind Configuration to Device Address* check box is unchecked then this text box will be enabled. In this text box you need to directly enter the value for the corresponding configuration. These configuration values should be within the Maximum and Minimum value of the currently selected AI channel. After entering the value if you click on the *Set* button then only that configuration is set.

If you select the RATE OF CHANGE configuration parameter ROC Text box will be enabled and the remaining text boxes will be disabled. This text box will take not only digital numbers you can enter float point values also. Once you entered ROC value you need to click on *Set* button then only this parameter is configured.
Note: *This ROC value should be in between 0.01 and 100.00.*
Note: *Once you configured these configurations parameters and switch to Auto Scale mode then you will lost the first five configurations. These configurations will be invalidated in Auto Scale mode. If you switched back from Auto scale mode to non Auto Scale mode you need to reconfigure (Basically Set) the parameters.*



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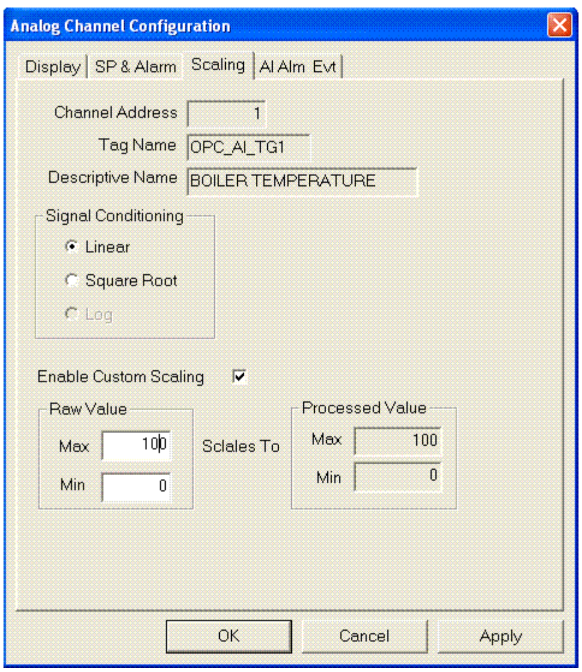
The Third Property Page in this Property Sheet is shown in below fig. With this property page you can set signal conditioning related parameters.

The first Signal Conditioning radio buttons, for ***Linear*** there is no change in the Process Variable (*PV or AI*) value. It will be used the same value that got from device. In ***Square Root*** mode it will square roots the incoming PV value. In the ***Log*** mode it will converts to logarithmic value.

In addition to above conversion you can make custom scaling by selecting the *Enable Custom Scaling* check box. It can allow you to select only in Non-Auto Scale mode only and you need to select the ***Linear*** Signal conditioning mode only.

In the Raw Values check boxes you can enters Input variable range those read from device, In the processed value you will get automatically Max and Min values those entered in the fist property page of this Analog Channel configuration. At runtime, every time acquiring the data this package automatically converts Raw Value to Process Value.



Example: One of your channels you are configured its ***Raw values*** are 30 and 120 and ***Processed value*** is defined as 300 and 1500 (*i.e., Maximum and Minimum values set in the first property page*). When you scan, if you got the PV as 50 (*i.e., Raw Value*) then it will be converted as 566 (approx) as process value.

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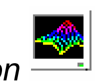
2.10 Alarm & Event Configuration:

Alarm & Event Configuration Property sheet consist of 4 property pages for each channel type (*AI, DI, AO & DO*). As per the process industry terminology you can monitor three kinds of events.

ALARM: As the PV is moving from Normal condition to abnormal condition is called as an Alarm.
EVENT: If the PV is recovering from abnormal state to normal state is called as EVENT.
MESSAGE: Any changes occur, you want to log those changes, you can configure those changes as MESSAGE. Ex: If you want to change any of the RO register or Activate any of the Coil of the connected device then you can log these changes as MESSAGE.

Users can display this Alarm & Event Configuration Property sheet in any one of the following methods:

- *By clicking on the menu item* Configuration → Alarm & Event Configuration.
- *By Clicking the* Analog Channel Configuration *tool bar button* 

For Alarm Beep related configurations, see the Chapter ***7.2 Printer & Alarm Beep*** related settings.

Analog Input (AI) Channel Alarm Configuration: Alarm configuration for AI Channel is a two step process.

- In this first stage it required to assign valid HIHIAlARM, HIALARM, LOALARM, LOLOALARM , ROC Alarm values for each from those you want subscribe alarms.
- In the second stage you need to subscribe the required alarm conditions.

For AI channels you can find the following Alarm Conditions. As PV (*Process Variable or Analog* Input *or Read Only Register Value*) Changing from:

- LT HI ALARM VALUE TO GT HI ALARM VALUE
- GT HI ALARM VALUE TO LT HI ALARM VALUE
- LT HI HI ALARM VALUE TO GT HI HI ALARM VALUE
- GT HI HI ALARM VALUE TO LT HI HI ALARM VALUE
- GT LO ALARM VALUE TO LT LO ALARM VALUE
- LT LO ALARM VALUE TO GT LO ALARM VALUE
- GT LO LO ALARM VALUE TO LT LO LO ALARM VALUE
- LT LO LO ALARM VALUE TO GT LO LO ALARM VALUE
- RATE OF CHANGE (ROC) ALARM

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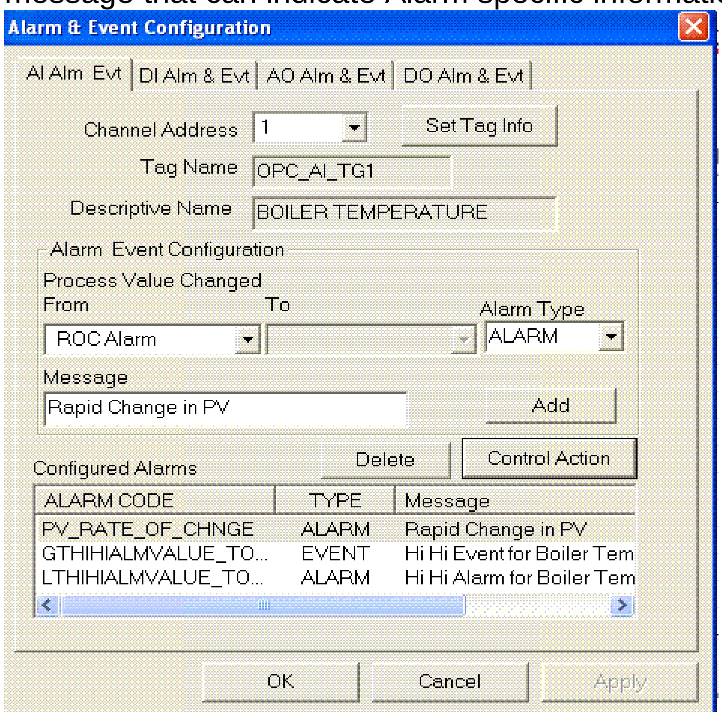
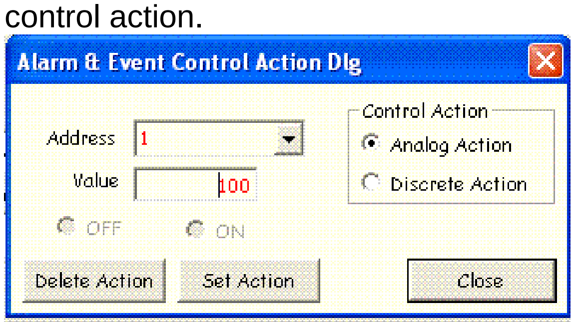
With the below AI Alarm & Event configuration property page you can subscribe any one or all of the above alarm conditions. At the time of subscribing the Alarm you can specify Alarm type can be any one of the following ALARM, EVENT or MESSAGE.

Once Channel Address selected all the configured alarms are populated to the bottom List control. Now you can subscribe new alarms or you can delete any subscribed alarms.

At the time of subscribing the Alarms you can optionally enter any message that can indicate Alarm specific information.



For each subscribed alarm you can specify Control Action. This control action will automatically perform whenever that particular alarm occurred. To define Control Action first you need to select subscribed alarm for which you want to define Control Action and just click on Control Action button. Then the following dialog will display to define control action.



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The control action can be Analog type or Discrete Type.
Analog Action: This type of control actions you can change any of the Analog RW register (*or Holding Register*) value. Ex: If any of the Hi Alarm occurred you can change any of the PID parameter values on the connected device.

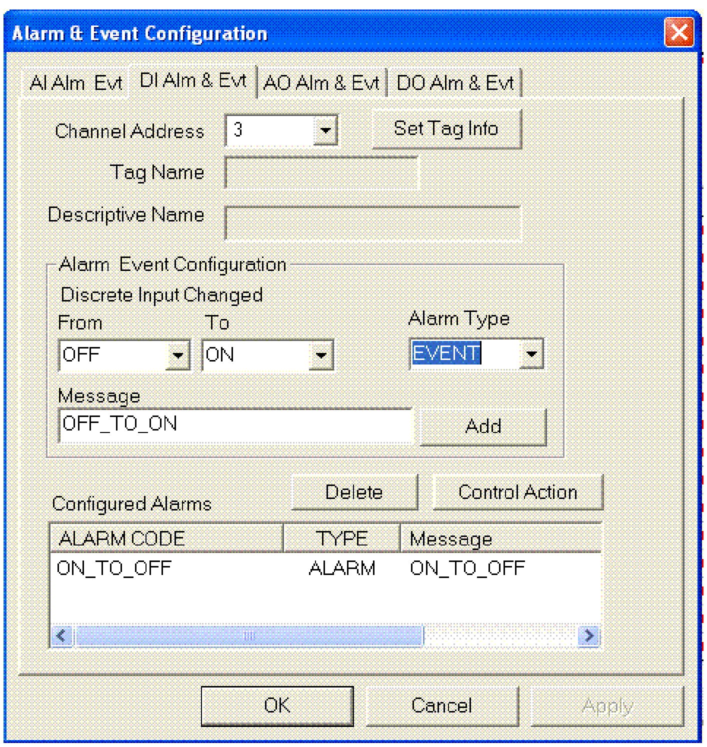
Discrete Action: For Discrete actions you can activate or deactivate any of the Output coils (*Discrete RW Register*). Eg: If any of the alarm occurred you can CLOSE the Fuel Valve. Whenever the Alarm recovered to Normal state (*i.e., EVENT occurred*) then you can configure to OPEN it automatically.

Discrete Input (DI) Channel Alarm Configuration: Unlike AI configuration this is just one step process. You can find following types of alarm conditions for DI Channels:

- ON TO OFF : *DI value changed from 1 to 0.*
- OFF TO ON : *DI value changed from 0 to 1.*

You can subscribe the above alarms conditions from the second property page of Alarm & Event Configuration Property sheet. These alarm conditions can be any one of alarm type ALARM, EVENT or MESSAGE.

DI Alarm & Event configuration property page is shown in below fig:



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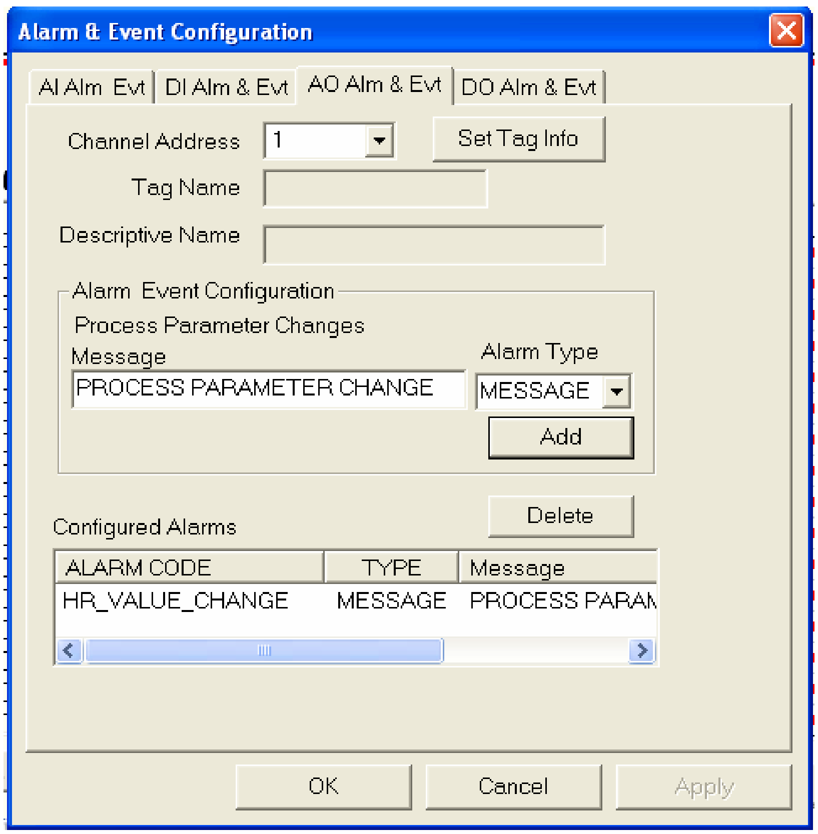
Like AI channels for each subscribed alarm conditions you can define Control Action and these control actions are either Analog action or Discrete Action. These actions are automatically executed whenever the alarm condition occurred.

Analog Output (AO) Channel Alarm Configuration: For AO channels (*Holding Registers*) also one step subscription for Alarm. For the AO Channel there is only one type of alarm condition is defined:

- HR VALUE CHANGE

Whenever manually changing the AO Channel values (*Holding Register*) values these alarms are generated. There is no Control Actions for these alarms. Alarm type can be any type of ALARM, EVENT or MESSAGE. Normally it will be type of MESSAGE.

AO Channel Alarm Configuration is shown in below fig:



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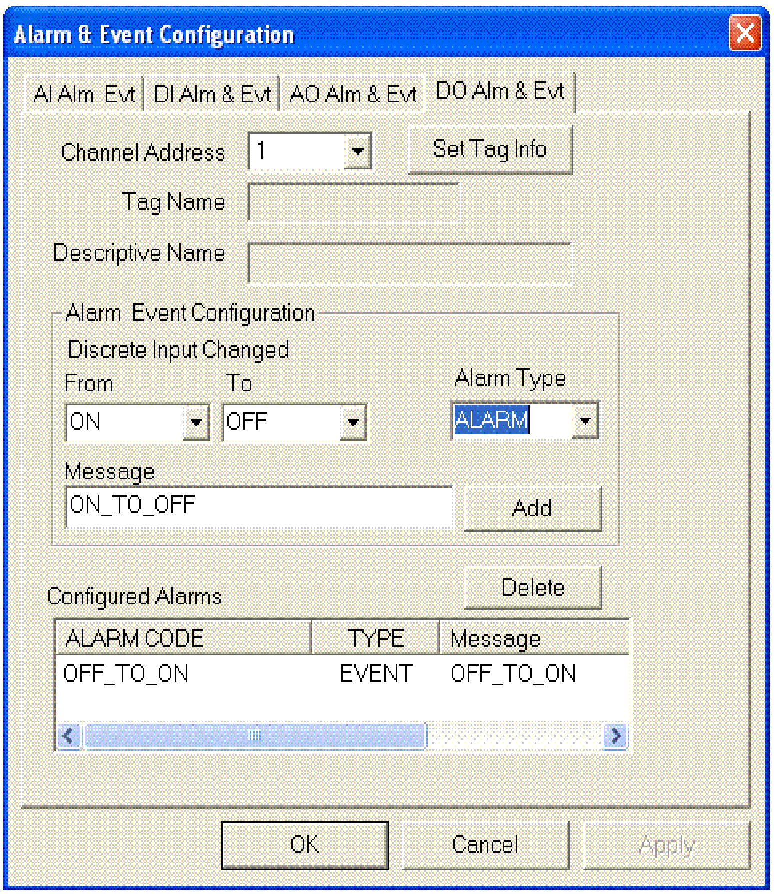
Discrete Output (DO) Channel Alarm Configuration: For DO channels (*Output Coils*) also one step subscription for Alarm. For the DO Channel the following types of alarm conditions defined:

- ON TO OFF : *DO value changed from 1 to 0.*
- OFF TO ON : *DO value changed from 0 to 1.*

Whenever manually changing the DO Channel values (Output Coils activated or deactivated) these alarms are generated. There is no Control Actions for these alarms. Alarm type can be any type of ALARM, EVENT or MESSAGE. Normally it will be type of MESSAGE.

DO Channel Alarm Configuration is shown in below fig:

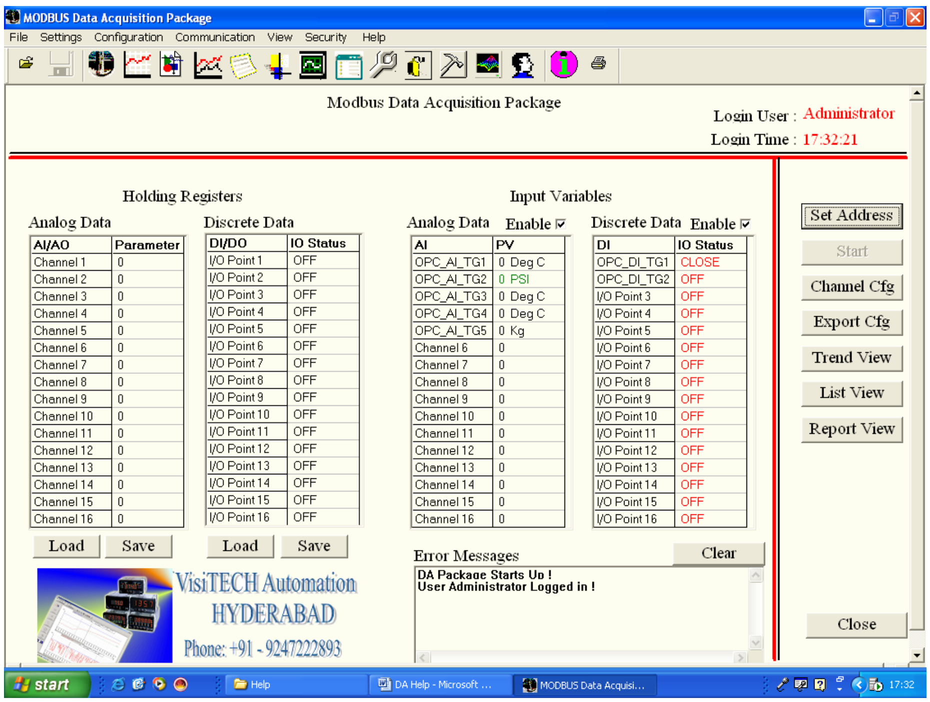


3. DA Views:

This Data Acquisition Package consists of 6 different kinds of screens. These screens will help to present and monitor the data in various formats. User can easily switch between different screens either by selecting the menu item or by clicking on buttons available in various screens.

3.1 Front Panel:

This is the first and major screen in this Data Acquisition package. All kind of channel parameters you can view from this front panel. The basic front panel screen is shown in below fig:



In this Front Panel you can see two different kinds of Channels. The left set of Channels is Read Write Channels (*i.e., AO and DO*) and the right set of channels is Read Only Channels (*i.e., AI and DI*).

At the right side of this Front panel there are set of buttons. By clicking on these buttons you can either pop up configuration dialogs are you can switch to other screens.

The *Start* button is initially disabled. Once you loaded communication driver and established the connection with device this button is automatically enabled. How to establish a connection with device you can see "***How to Communicate with Device"*** chapter.

With the *Export Cfg* button you can export the device and communication configuration parameters to user specified text file.

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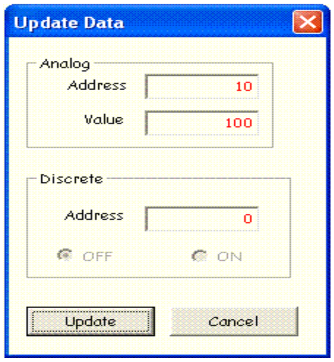
By clicking on the bottom *Close* button shut down's the application.

In this front panel at upper right corner displays the *Login User Name* and *Login Time*. How to Add or Remove users please see *Login User Management* Chapter.

With *Load & Save* buttons you can load the RW Register (*Both Holding Registers AO and Output Coils DO*) values from the connected device registers and you can save them to device registers.

To change any of the register value just double clicking on the corresponding channel an *Update Data Dialog* will pops up. In this dialog you can change the values and finally clicking on the *Update* button it will automatically updates the corresponding channel value.



Finally you can save this value to the Device Registers by clicking on the *Save* button. Before saving the values to the Device it required to need to establish the connection with device.

Once you established the communications with devices and started acquiring the data, the data will be automatically updated in the right side RO (*Both AI & DI)* channels. These channels data will be updated in each scan.

If you want to scan only any one kind of channel per moment you can clear the other type of channel *Enable Check box.* Once it cleared the corresponding channels data is not going to be read from the device until it will be checked.

If any of the AI Channel value crosses the alarm limits then the color of the corresponding channel value will be changed to different color.

If the PV is above the HiHiAlarm or below the LoLoAlarm value then color will be appeared as *Red*. Similarly if the PV is above the HiAlarm value and below the HiHiAlarm value (Similarly Below the LoAlarm value and above the LoLoAlarm Value) then the PV is displayed as *Green*. Within the normal range it will be displayed as *Black*.

If no alarms are configured for any particular channel then the corresponding channel value always displayed as *Black*.

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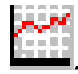
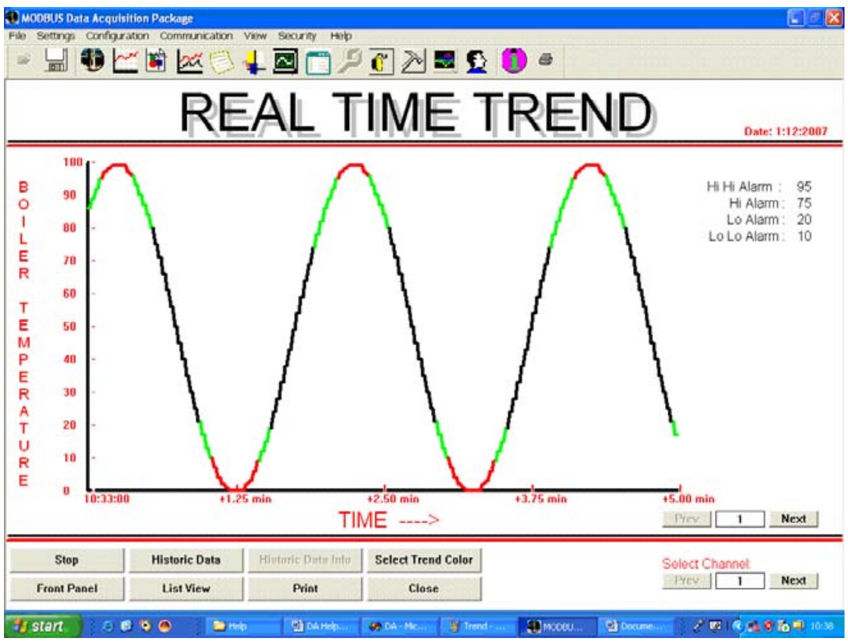
3.2 Trend View:

In this Data Acquisition package we have three different types of Trend Views. Each trend view have different features. This is the first and simplest trend view. User can easily switch to this trend view either any one of the following methods.

- *By clicking the* Trend View button *on the Front panel or any other screen or view.*
- *By selecting the menu item* View →Trend View*.*
- *By clicking the Trend View tool bar button* *.*

The trend view diagram is shown in below fig:



This Trend View always displays X-Axis as Time and Y-Axis as any one of the Analog Channel. User can easily switch between channels by clicking *Next* and *Prev* buttons under Selected Channels.

If the selected channel is in *Auto Scale* mode the Y-Axis *Max, Min values* are automatically updated while acquiring the real time data.

This Trend view displays the trend for Real Time Channels and Historic channels. Users can easily switch between Historic and Real time trends by clicking on the *Historic Data* button at the bottom of the Trend View screen.

For a particular channel if you are configured Alarm values and PV crosses the alarms limits trend colors will be changed. By clicking the

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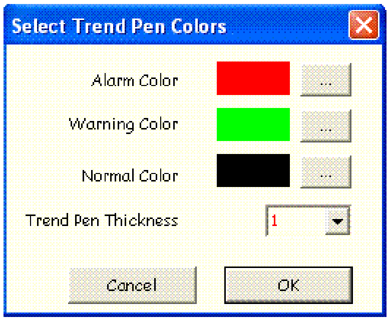
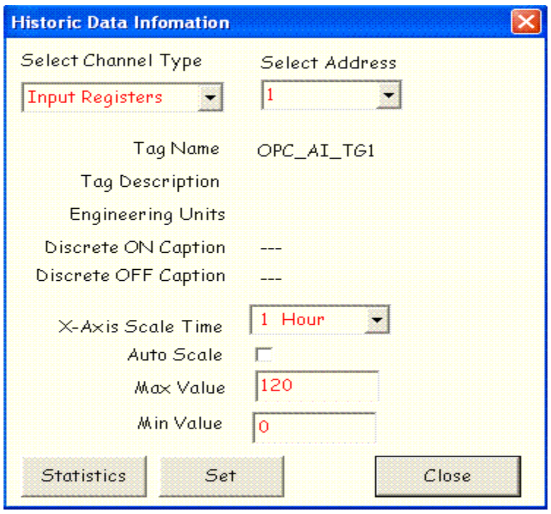
Trend Color button the following dialog displays in this dialog user can select different trend colors for Alarm, Warning and Normal values of PV. You can also change the Trend pen thickness for the trend. If Alarm values complete trend displays with normal color. These color are used for all channels in this trend screen.



Users can change the Time Scale period for Real time channels by selecting the appropriate radio button in the Analog Channel Configuration property page. For historic channels by clicking on the *Historic Data Info* button the following dialog will displays. In this dialog users can change Max, Min values and X-Axis Scale Time by clicking the *Set* button the new values will be updated on the Trend screen.



Within this you can see the Tag Name, Descriptive Name, Eng Units for AI channels. Similarly for DI Channels you can see Tag Name, Descriptive name and ON, OFF captions.

Any changes in this Historic Data Information will be valid only for that session. These changes are not saved into the file.

In this trend screen there couples of buttons at the bottom of the screen with these buttons users can easily switch to other screens or close the applications. To print the current trend use *Print* button.

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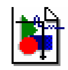
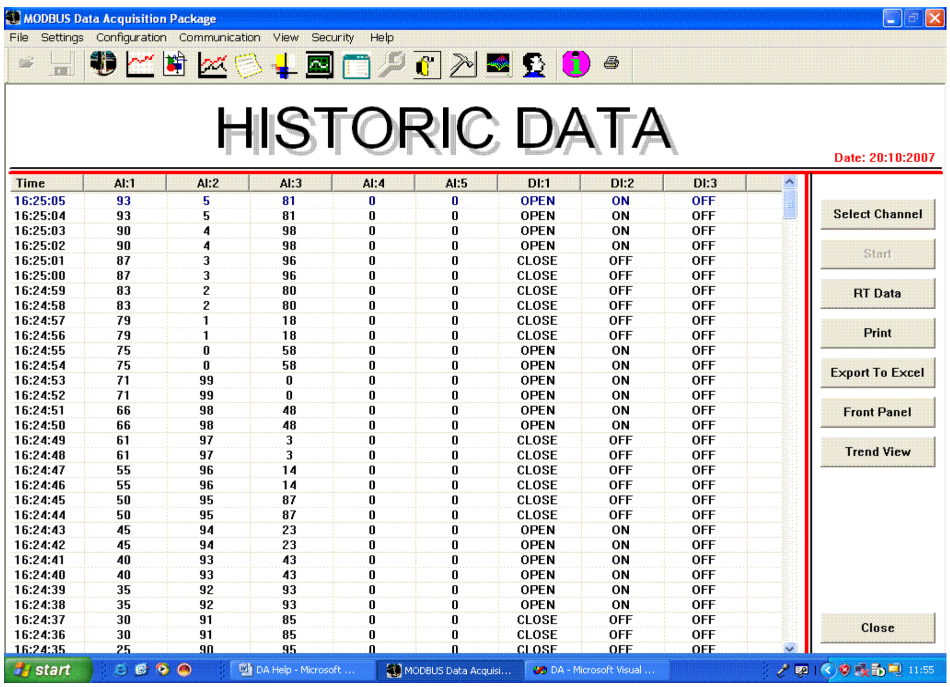
3.3 List Data View:

In this View you can see the AI & DI Channels data with time stamp. This view can displays Real time as well as Historic data. User can easily switch to this *List View* either any one of the following methods.

- *By clicking the* List View button *on the Front panel or any other Screen or View.*
- *By selecting the menu item* View →List Data View*.*
- *By clicking the* List View *tool bar button* 

The List view screen is shown in below fig:



In this view, left side List control displays the configured data and right side there are set of buttons. In this view you can see the AI and DI channels. Also you can configure to see only user interested channels. To configure the channel by clicking on *Select Channel* button Select Channel dialog will be displayed. In this dialog Users can select only user interested to channels to display.

The Channel Selection dialog is shown in below fig. This dialog used to select for both Analog & Discrete Channels. In this dialog you can see the Channel Tag Name (*If configured*) other wise it display the channel address.

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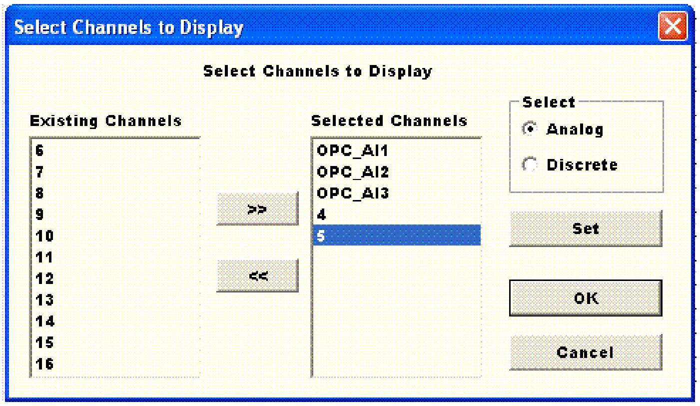
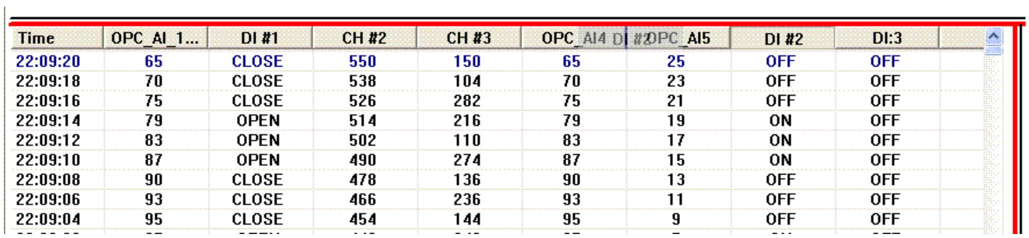


Once user selected the channels, user can position the channel at user specific location. Just by clicking the header button (*displays the Tag Name or Address of channel*) and drag it near to the user interested position.



Time	OPC_AI_1...	DI #1	CH #2	CH #3	OPC_AI4	OPC_AI5	DI #2	DI:3
22:09:20	65	CLOSE	550	150	65	25	OFF	OFF
22:09:18	70	CLOSE	538	104	70	23	OFF	OFF
22:09:16	75	CLOSE	526	282	75	21	OFF	OFF
22:09:14	79	OPEN	514	216	79	19	ON	OFF
22:09:12	83	OPEN	502	110	83	17	ON	OFF
22:09:10	87	OPEN	490	274	87	15	ON	OFF
22:09:08	90	CLOSE	478	136	90	13	OFF	OFF
22:09:06	93	CLOSE	466	236	93	11	OFF	OFF
22:09:04	95	CLOSE	454	144	95	9	OFF	OFF

For example if any of the DI input value depends on AI value then putting those AI & DI channels together user can easily monitor the status of the DI. Above fig, user trying to drag the DI# 2 channel to position it in between OPC_AI4 & OPC_AI5 channels. Similarly DI# 1 discrete channel is dropped near to the OPC_AI_1 analog channel.

For Real Time List view, at data acquiring the data will be updated to this view every time scan of the data.

With the help of *Start* button you can start and stop acquiring the data. This button is enabled once connection with device is established. How to establish a connection with device you can see "***How to Communicate with Device"*** chapter.

With the help of Print button user can print the user selected channels data only (*Not all channels data is printed*). If user selects all the Analog and Discrete channels then it displays and prints the all channels.

With the help of *Export To Excel* button, user can export the data of the selected channels to user specified file in the "csv" format. These "csv" files can be read by Microsoft Excel and easily converted into Excel documents for further processing.

Below buttons helps to switch the views and bottom *Close* button will shutdowns the applications.

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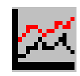
3.4 Multi Channel Trend View:

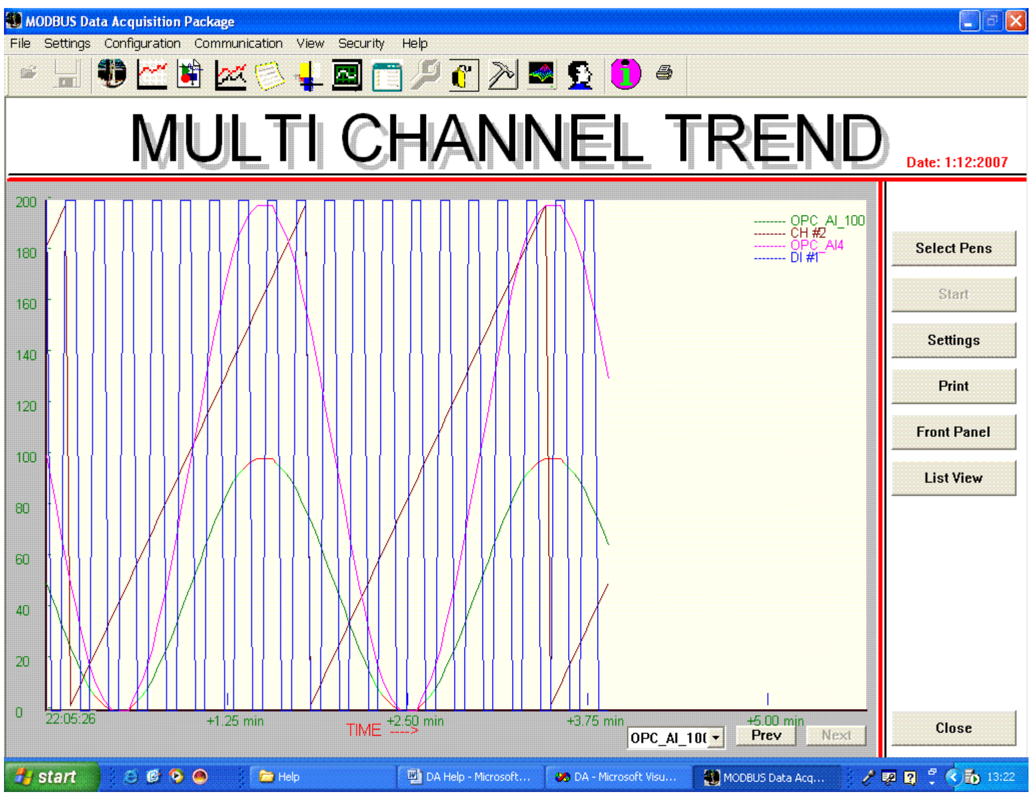
This trend view is second type of trend view in this Data Acquisition Package. This trend view consists of two special features compared to Normal trend as specified in 3.2 Trend view chapter.

1. First it displays the multiple channels (*Maximum of* 4) in the same view. These channels can be either super imposed or separate traces.
2. In addition to AI Channels it can also displays DI Channels.

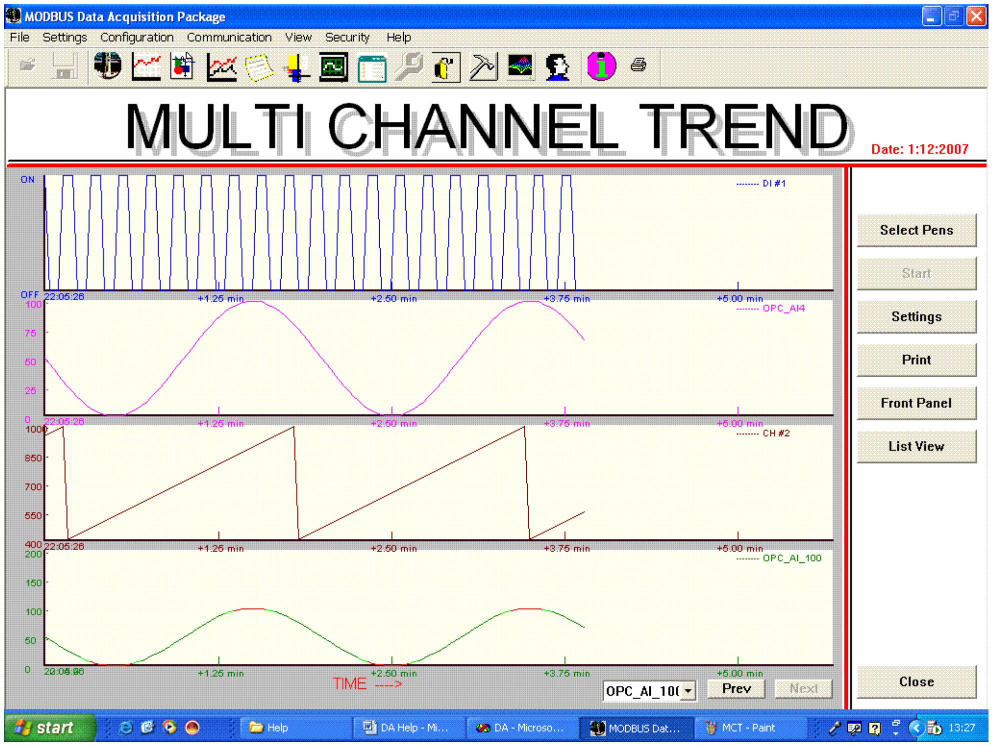
User can easily switch to this *Multi Channel Trend View* either any one of the following methods.

- *By selecting the menu item* View →Multi Channel Trend View.
- *By clicking the* Multi Channel Trend View *tool bar button* 

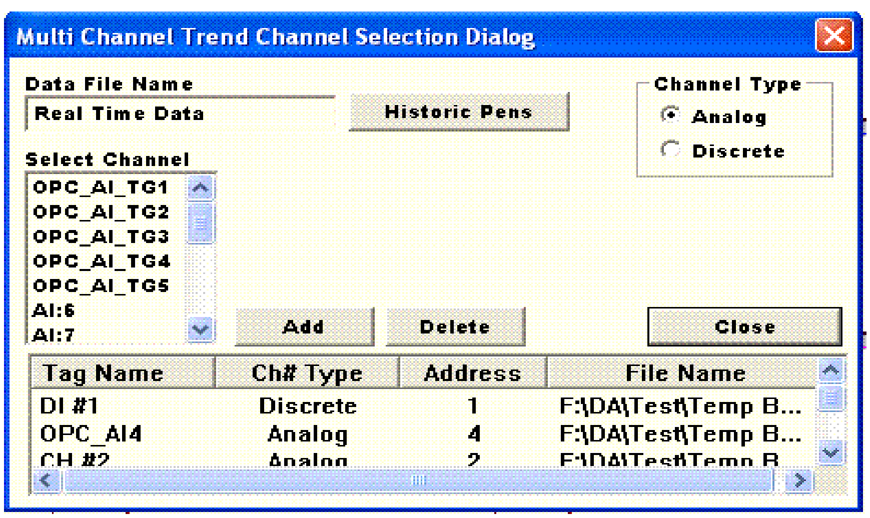
The Multi Channel Trend screen is shown in below fig:



This Multi Channel Trend screen will display the trend in two different modes. They are *Super imposed mode* or *Separate traces*. The above fig is the display of super imposed mode. The below fig display the separate traces of the same trend channels.



These trend channels can be either Real Time or Historic or combination of both. In the case of Historic you can select the channels from four different files (*four different batches data*) also. Once selected the trend channel, initially different color for different channel is assigned. User can change the trend channel to different color in the *Settings* dialog.



By clicking on the *Select Pens* buttons Channel Selection Dialog appears. In this channel selection dialog you can select the channel from Real Time and Historic data, also both AI & DI channels. The

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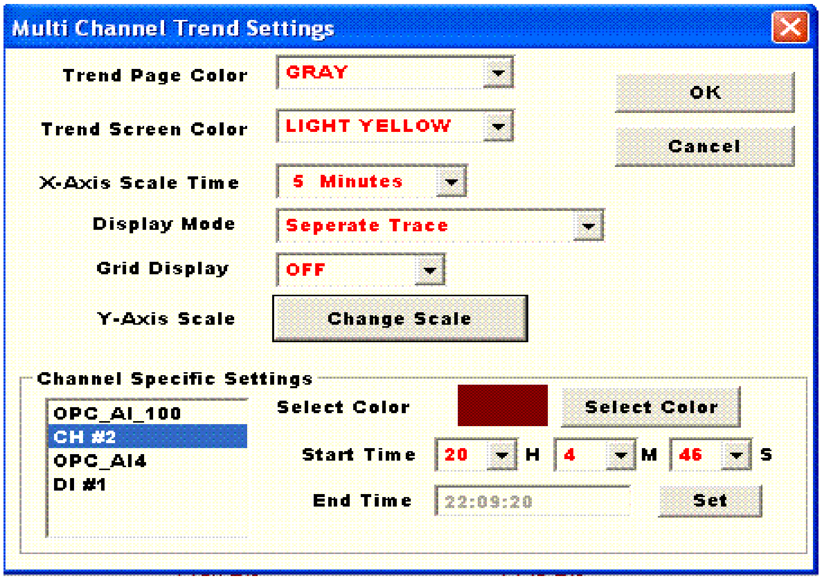
Channel Selection dialog is shown in above fig. In the above dialog user can select the historic data by clicking on *Historic Pens* button. User can easily switch Analog and Discrete channels by clicking on the appropriate radio button.

Once Historic Data or Channel type selected, immediately *Select Channel List Box* updates the corresponding channels automatically. By selecting the channel in this list box you can add the channel. User can also delete any of the selected channels simply by selecting that channel in the bottom list control and clicking on *Delete* button.

Once trend channels are selected and displayed then you can change its appearance, Time scale values, Max/Min values, Trend pen colors, etc., of each channel by clicking the Settings button. If any changes in this Settings dialog will effects only to this Multi Channel Trend View only.



At the bottom of this trend screen, you can find a *channels* Combo box and *Next, Prev* buttons. This channel combo box displays all the selected channels. Whatever the channel selected in this combo box is called active channel. Whenever user clicks on Prev or Next button that particular channel trend only moves.

This Multi channel trend is used to study the any particular channel values in various batches.

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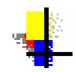
3.5 Alarm & Event View

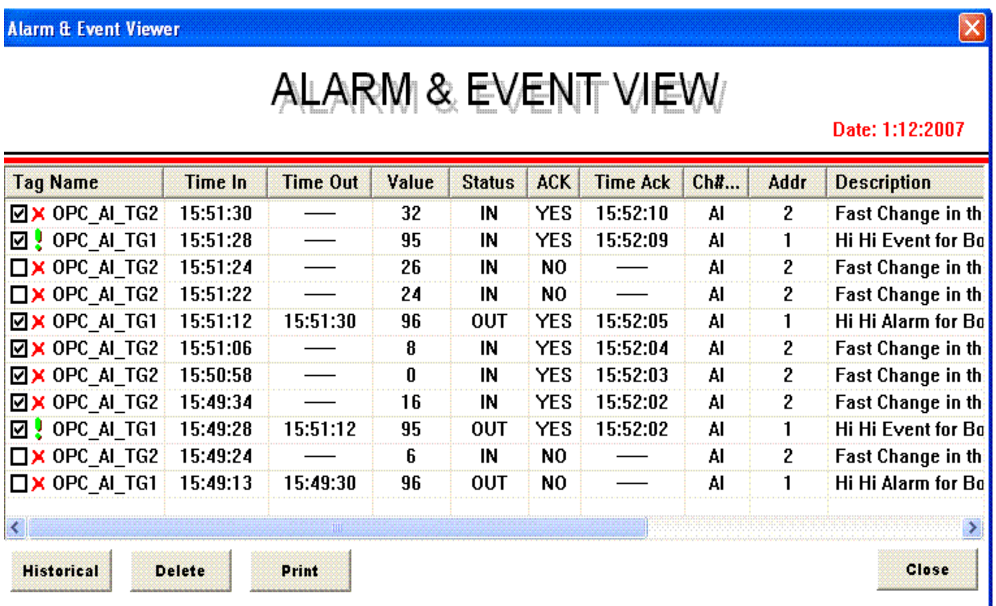
This Alarm & Event view is one of the most useful view. With this view you can see the Real time (*i.e. today*) Alarms & Events and Historic Alarms & Events.

Every day Alarm & Event data is automatically stored in a separate file in the *Alarm* directory within the Data Acquisition Package executable directory. All these files will be stored with extension of "*EVT*".

User can easily display this Alarm & Event view in either any one of the following methods.

- *By selecting the menu item* View →Alarm & Event View*.*
- *By clicking the* Alarm & Event View *tool bar button* 

The Alarm & Event View is shown in below fig. This Alarm & Event view is not a full view, while this Alarm & Event view is displaying you can easily switch to different views and you can monitor other views while monitoring this Alarm data.



Tag Name	Time In	Time Out	Value	Status	ACK	Time Ack	Ch#...	Addr	Description
OPC_AI_TG2	15:51:30	—	32	IN	YES	15:52:10	AI	2	Fast Change in th
OPC_AI_TG1	15:51:28	—	95	IN	YES	15:52:09	AI	1	Hi Hi Event for Bo
OPC_AI_TG2	15:51:24	—	26	IN	NO	—	AI	2	Fast Change in th
OPC_AI_TG2	15:51:22	—	24	IN	NO	—	AI	2	Fast Change in th
OPC_AI_TG1	15:51:12	15:51:30	96	OUT	YES	15:52:05	AI	1	Hi Hi Alarm for Bo
OPC_AI_TG2	15:51:06	—	8	IN	YES	15:52:04	AI	2	Fast Change in th
OPC_AI_TG2	15:50:58	—	0	IN	YES	15:52:03	AI	2	Fast Change in th
OPC_AI_TG2	15:49:34	—	16	IN	YES	15:52:02	AI	2	Fast Change in th
OPC_AI_TG1	15:49:28	15:51:12	95	OUT	YES	15:52:02	AI	1	Hi Hi Event for Bo
OPC_AI_TG2	15:49:24	—	6	IN	NO	—	AI	2	Fast Change in th
OPC_AI_TG1	15:49:13	15:49:30	96	OUT	NO	—	AI	1	Hi Hi Alarm for Bo

In this Alarm & Event view you will find the set of columns. The description of these columns is shown below fig.

Tag Name: In this column you will find three different types of information:

- Tag Name of the Alarm Source Channel.

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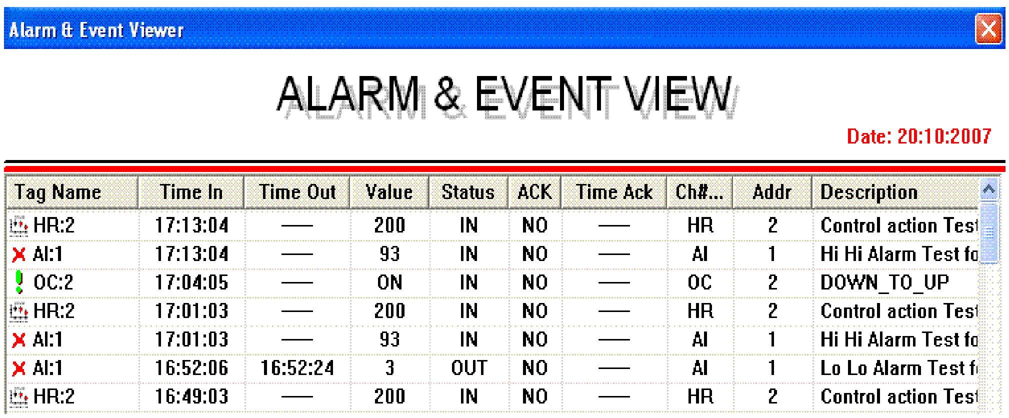
- Alarm Type (*i.e., ALARM, EVENT or MESSAGE*). Alarms are indicated with , Similarly Events are indicated with , and Messages are indicated with .
- The check boxes are used to acknowledge the alarm. After generating the alarm whenever the operator acknowledges he just simple clicks on it. The acknowledge time will be logged. In the above view you will see the acknowledge time in the 7th column. This check box is not visible for Historic Alarms & Events (*i.e., Only visible for Real Time Alarm & Event*). You can see the below is snap shot of Historic Alarm & Event View.


Alarm & Event Viewer

ALARM & EVENT VIEW

Date: 20:10:2007

Tag Name	Time In	Time Out	Value	Status	ACK	Time Ack	Ch#...	Addr	Description
HR:2	17:13:04	—	200	IN	NO	—	HR	2	Control action Test
AI:1	17:13:04	—	93	IN	NO	—	AI	1	Hi Hi Alarm Test fo
OC:2	17:04:05	—	ON	IN	NO	—	OC	2	DOWN_TO_UP
HR:2	17:01:03	—	200	IN	NO	—	HR	2	Control action Test
AI:1	17:01:03	—	93	IN	NO	—	AI	1	Hi Hi Alarm Test fo
AI:1	16:52:06	16:52:24	3	OUT	NO	—	AI	1	Lo Lo Alarm Test fi
HR:2	16:49:03	—	200	IN	NO	—	HR	2	Control action Test

In the above screen short thee is no acknowledgement check boxes in the first column. The historic data date will be displayed at the upper right corner of this view.

Time In: This column display the time when the alarm is generated.

Time Out: If the alarm is recovered (*i.e., the PV comes from alarm value range normal value*) then this column displays recovered time.

Value: Whenever the alarm generated the corresponding PV value will be displayed in this column.

Status: This column indicates the corresponding channel is in Alarm area or recovered.

ACK: The operator acknowledged time for that particular alarm.

Time ACK: Time of acknowledgement of that particular alarm.

Ch# Type: This column displays the alarm source channels type (*i.e., AI, DI, AO or DO*).

Address: This column displays the alarm source channels address.

Description: Description of the alarm.

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At the bottom of the Alarm & Event view you will find the following buttons:

Historical: By clicking on this button you can switch between Real Time and Historic Alarms displays. Whenever switching to Historic display it prompt for Historic data file.

Delete: This button is enabled if the currently login user has Administrative privileges. (*You can see more about Administrative privileges in* Login Management Chapter). This button is used to delete any of the alarms in the alarms list.

Print: This button is used to print the alarms.

Close: By clicking on this button Alarm & Event view is closed.

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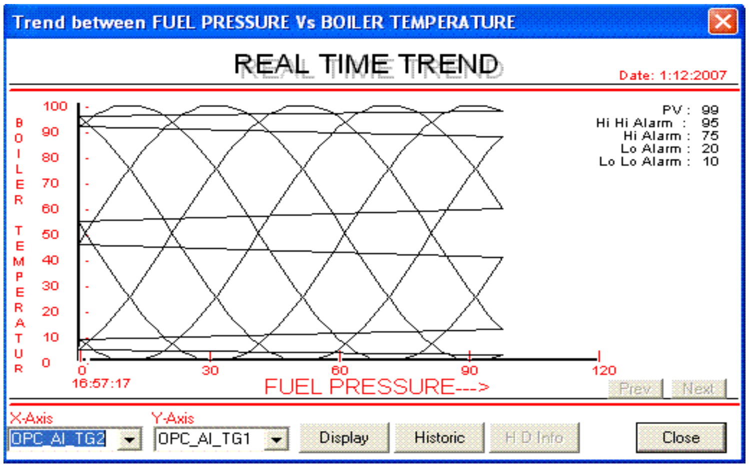
3.6 Child Trend View

This is the third type of trend view in this Data Acquisition Package. This trend view occupies only small part of screen. At a time you can view four this type of trend views.

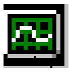
This trend view has the following special features:

- In this trend view, for X-Axis not only *Time* but you can assign any one of the AI channels. For Y-Axis as usual any AI channel. The below diagram represents the trend between two different AI channels.



- It displays only AI Channels. (*No DI channels as in Multi Channel Trend*).
- At a time maximum of 4 Child trend screens you can view and each trend can represent Historic data from four different data files or any Real Time channel.
- While this Child trend views are displaying user can easily switch between other view. So user can simultaneously monitor over trend and data views simultaneously.

User can easily display this Child Trend view in either any one of the following methods.

- *By selecting the menu item* View →Child Trend View.
- *By clicking the* Alarm & Event View *tool bar button* 

The below diagram represent the multiple Child Trends while watching the List Data View.

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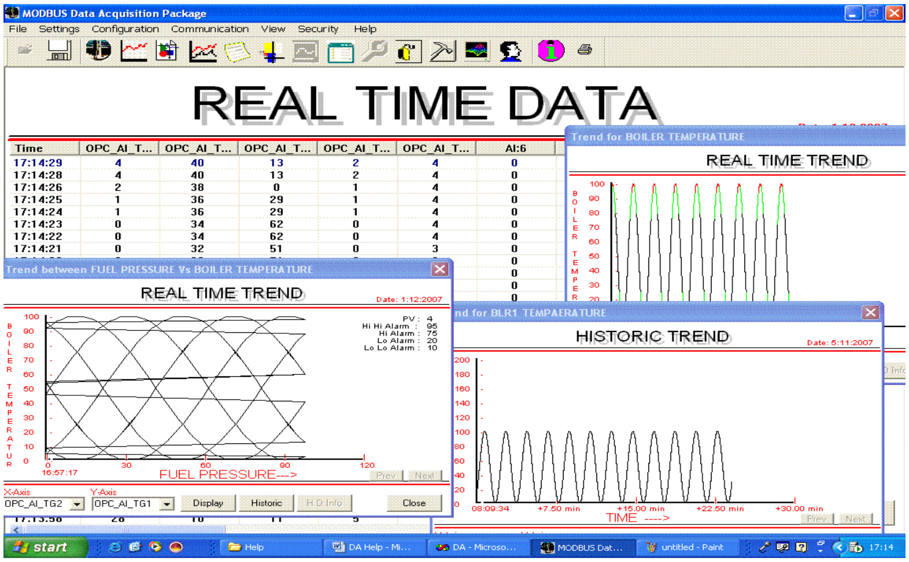

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4. How to Communicate with Device

4.1 Connecting to Device:

To establish communication with device take the following steps:

- First select the communication protocol in the *Device & Communication's* property sheet's *Protocol Configuration* property page. While selecting the communication protocol observed **Communication Medium** type. To access the Device & Communication Property sheet just select the menu item *Configuration → Device Communication Settings* or select the *Device & Communication* Settings tool bar button.
  - ➢ If the communication medium type is SERIAL just switch to *Serial Port configuration* property page in the same property sheet and select the DA device connected serial port and make appropriate settings. For device serial settings you can contact filed engineer.
  - ➢ If the communication medium type is ETHERNET just switch to *Ethernet Port configuration* property page in the same property sheet and assign the IP address and TCP/IP port no of the connected device.
  - ➢ If the communication medium type is OPC Component just switch to *OPC Server configuration* property page and get the available OPC servers on Remote server machine and select your OPC server check for availability. If OPC items are not configured, configure the OPC items.
  - ➢ If the communication medium type is OTHER, you don't need to configure any communication related settings just close this *Device & Communication* property sheet.
- Now you need to load the communication driver and required to connect to the device. To establish connection with device select the menu item *Communication →Connect.* Once it successfully established the connection with device you will see a message in Error Message list box as *"Successfully Connected !"* If any errors occurred those error messages will be displayed.
  - ➢ If Successfully connected now you can upload the AO & DO Parameters. If required change those values and save them back to the device.
- To start acquiring the data click on *Start* button. This button is available in all view. Just clicking on any view's *Start* button it will starts acquiring the AI & DI data.

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4.2 Disconnecting to Device:

- To disable the connection with device first stop acquiring the data by clicking the *Stop* button in any of the views. If you are not in acquiring mode you don't find the *Stop* button just switch to next step.
- Now, just click on *Communication →Disconnect* menu item. It will un-loads the communication driver and disable the communication with device. If the communication is not established then this menu item is grayed.

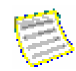
Note: *The menu items "*Communication→Connect" & "Communication → Disconnect" *are enabled when the active view is Front Panel. In all other views these menu items are disabled. To establish or disconnect a connection with device switch to front panel and take appropriate action.*

5. Report Generation Wizard:

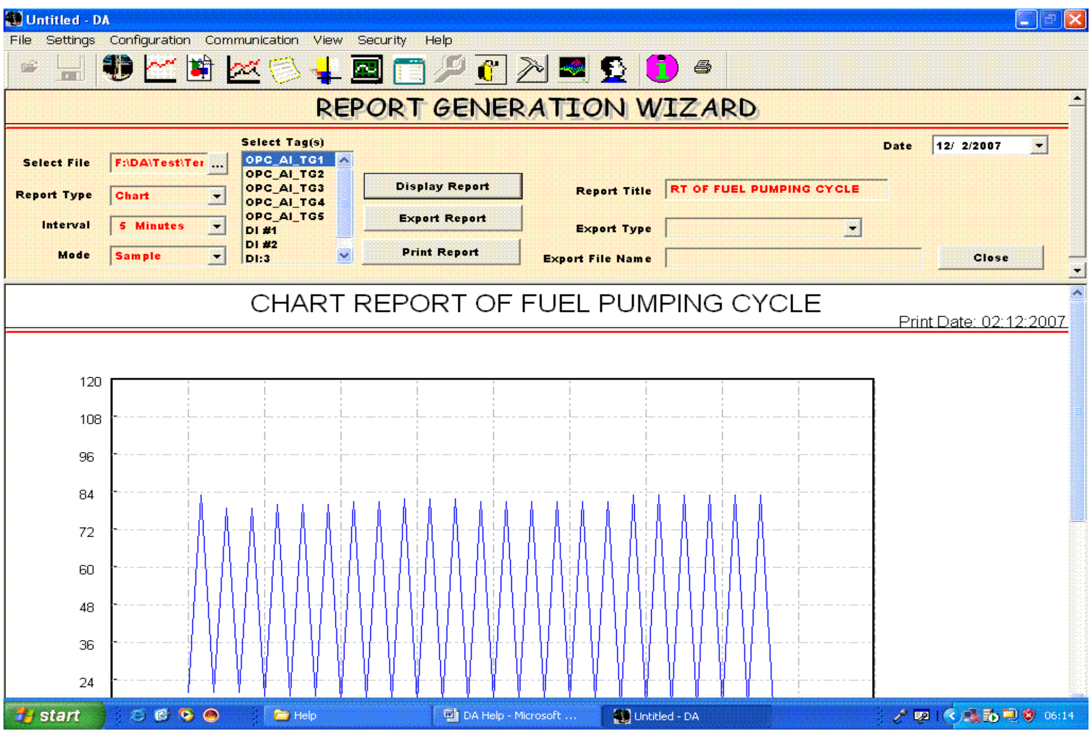
This Report generation wizard reads the data from DA Package generated data files and displays the reports in three different modes. They are CHART, CROSS TAB and DAILY reports. In addition to displaying the reports it will export the reports into various file formats.

With this wizard user can generate the reports for AI & DI channels.

User can easily switch to this *Report View* either any one of the following methods.

- *By selecting the menu item* View →Report View.
- *By clicking the following* Report View *tool bar button* 

This Report Generation Wizard consists of two different views as shown in below fig. The upper view is a dialog it will allow to select various report generation parameters and the bottom is report display view. All types of reports will be displayed in bottom view.



Select File: With this button you can select the Historic data file to generate the reports. Once the file is selected available channels are automatically populated into *Tags List box*. This list box holds the tags for both AI & DI channels.

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Report Type: In this combo box you will select the report type to generate and display (*export or print also*) the reports. At present this package is supporting three different types of reports they are CHART, CROSS TAB and DAILY reports. More details on these reports see the below sections in this chapter.

Interval: This combo box allows you to select the data sampling interval to collect the data from the data file to generate the report. The available sampling intervals are 1, 5, 10, 15 & 60 Minutes.

Mode: This column allows you to select the mode of collecting data. The available modes are Sample, Average, Maximum & Minimum.

- In the *Sample* mode it will simply take the process data at user specified intervals.
- In the *Average* mode it will averages the data at user specific interval.
- At *Maximum* mode it will find the maximum values in the user specified intervals. This *Maximum* mode will help you to find the worst conditions in the selected batch.
- At *Minimum* mode it will find the minimum values in the user specified intervals. This *Minimum* mode will help you to find the worst conditions in the selected batch.

Report Title: In this edit box you can enter the report title. This title will be display at the beginning of the report in report view or exported report.

Report Export Type: At present this package supporting three different formats of exporting the report. They are *Comma Separated Value (CSV)*, *Rich Text Format (RTF)* and *Text formats*. CSV format files are readable by Microsoft Excel from this excel you can convert into other formats also. You can't export the Chart report.

Export File Name: To specify the export file name. All the CSV format files should have csv extension, similarly all RTF files should have extension of rtf finally text files should have an extension of txt.

Date Control: This control will allow you to select the report generation date. This data you can see as print date in your report view.

Display, Export & Print Report: These buttons displays, exports and prints the report.

Close Button: By clicking on the Close button it will closes the report generation wizard and switches to Front panel.

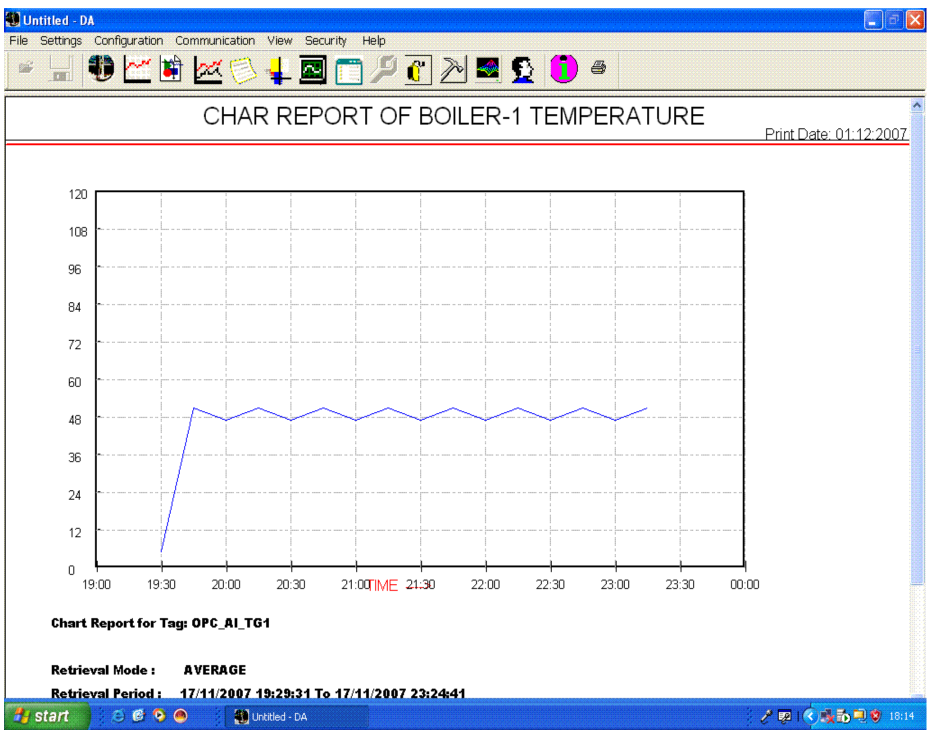
5.1 Chart Report:

This chart report gives the clear idea about the process data in the selected batch or data file. Within this report in Maximum and Minimum mode you can easily find out peek values in the given data. This report helps to study or gives the feed back of the selected channels data. You get different chart reports for different modes of selected for the same data for a selected channel.

The below figure is Chart report. This report is one of the best report to get the clear idea to get a channels data in a particular batch. In Max/Min values mode you will find any abnormal conditions exists in the given batch.

This report is used to submit the report to the higher authority at the end of the batch for any particular PV or Channels.

This Chart report generates the report only one channel at a time.



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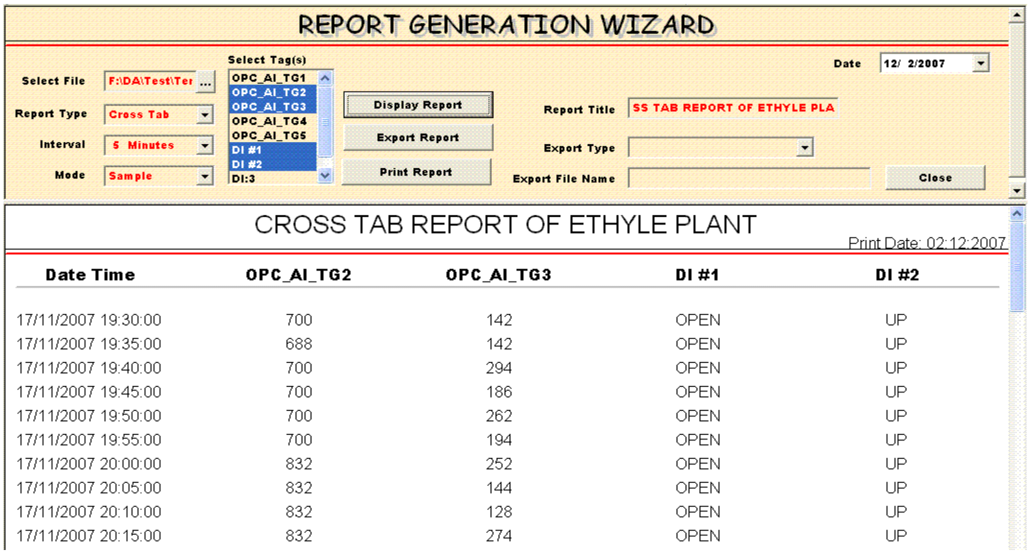
5.2 Cross Tab Report:

This Cross tab report is suitable to study or to perform comparative analysis between channels (*Both AI & DI*).

This Cross tab report generated Maximum 5 channels report at a time. If you have more channels you have to make multiple reports by selecting different channels.



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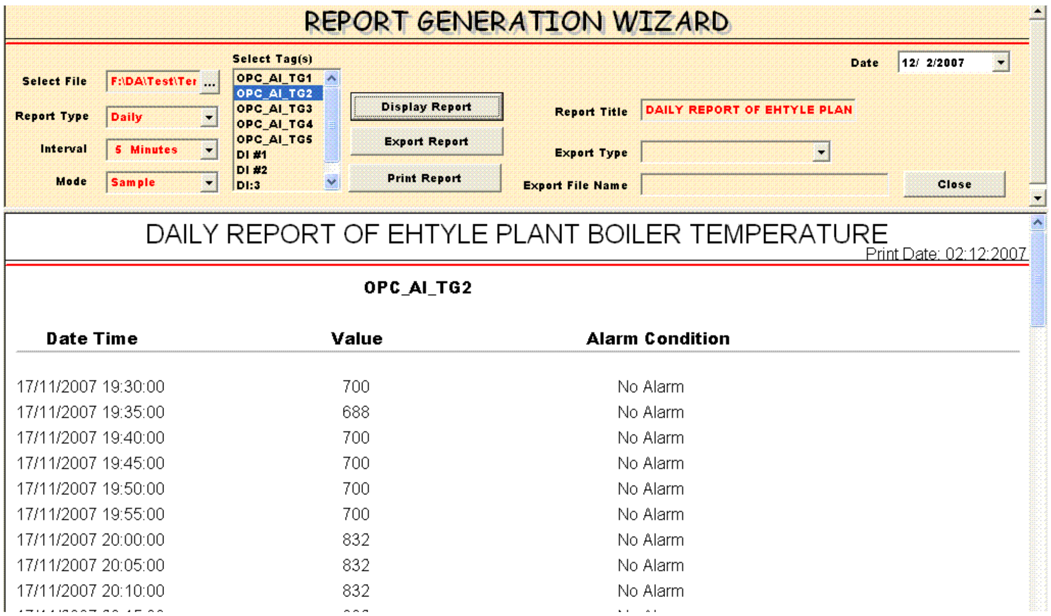
5.3 Daily Report:

This Daily report is best suitable to submit the daily reports for each channel at the end of the batch. At a sampling interval it will clearly indicates the PV value and Alarm condition status.

To get the report you can select any number of channels in a selected batch (or data file).



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5.4 Export Report:

In addition to displaying the reports you can Export the reports in various formats in users specified files. At presently this package is exporting the reports into the following formats:

- Comma Separated Value
- Rich Text Format
- Text Format

You can export only Cross Tab and Daily reports. You can't export Chart reports into file.

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6. Login Management:

6.1 User Management :

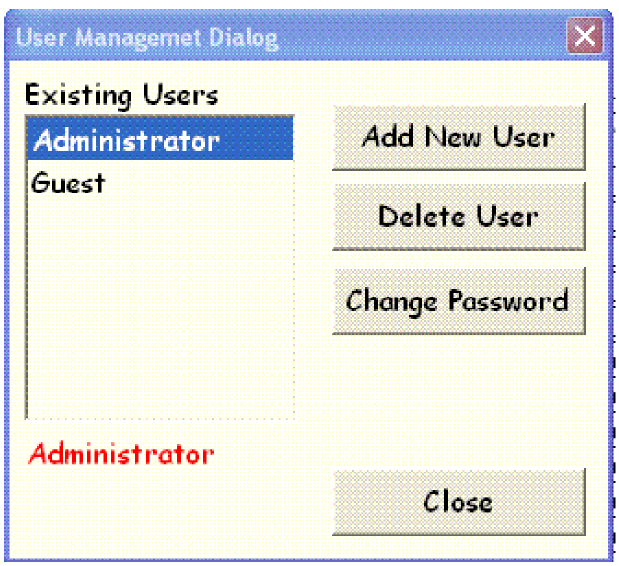
With the help of User Management dialog you can Add new users, Delete existing user and change the password.

You can access the User Management Dialog in any one of the following methods.

User can easily display this *User Management Dialog* in any one of the following methods. You can access this dialog from any view.

- *By selecting the menu item* Security →User Management*.*
- *By clicking the* following User Management *tool bar button* .

The following diagram shows the *User Management* dialog. In this dialog all the existing users are display in the left side list box. There will be two different types of users. First type of users are Administrative users others are Simple users. Administrative users have some special privileges those are explained *Administrative Privileges* section in this chapter.

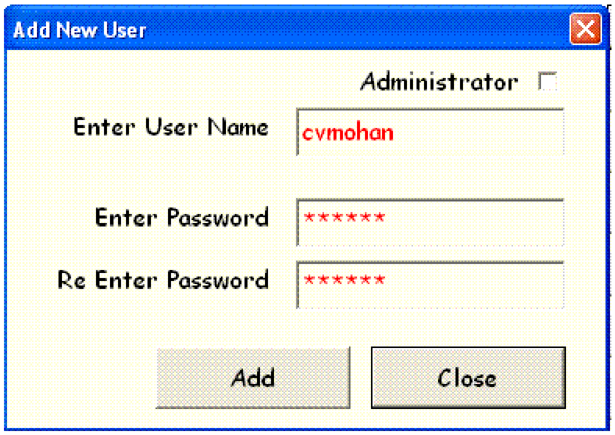


Whenever this package is installed and first time started the application the *Administrator* and *Guest* user are automatically created and added to the existing users list and they are called as system account users. In these uses Administrator has Administrative privileges and Guest is a Simple user. More on Administrative privileges see the below section.

The user of this package can't delete these system account users but they can change their default passwords.

- *Default password for the Administrator is* AUTOMATION.
- *Similarly default password for Guest is* 123.

- To add new user click on Add New User button in the User Management dialog. Once clicked the following displays to add new users.



If the new user required an Administrative privileges click on Administrator check box, then enter user name and password boxes then click add. Once the above dialog box is closed the new user will be appeared in the existing users list.

- To delete an existing user just select the user to delete in the users list and click on delete button. Take note system account (*i.e., Administrator & Guest*) users can't be deleted.
- To Change password select the user in the existing users list and click on Change password. More details on change password see the Change Password section in this chapter.

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6.2 Login User Change:

Once package is started up and event acquiring the data also you can change to different login. To change the login you don't need to restart the package. You can see the new login user and login time at the upper right corner of the front panel.

User can access the Change Login by clicking on *Security → Change Login* menu item. Once user selected the above menu item the following *DA Package Authentication* dialog displays, in this dialog by selecting the new user name, entering appropriate password and by clicking on Login, the new user login takes place.

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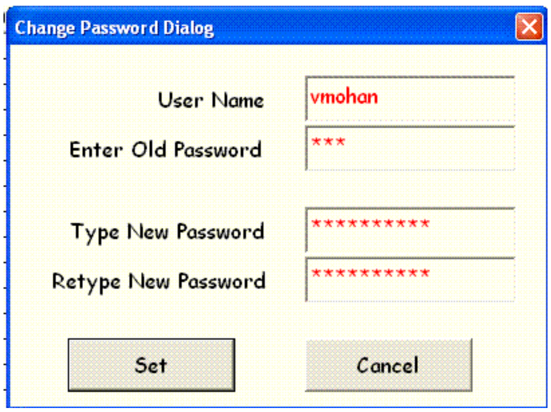
6.3 Change Password:

You can activate the Change Password dialog in any one of the two ways:

- By selecting the menu item *Security→Change Password.* Here user can change only currently login user's password.
- The second method is to display the User Management dialog and select any user wants to change the password and then click on Change Password button.

The below diagram shows Password change dialog here user name can't be changeable. Only users required to enter existing password and new password only.



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6.4 Administrative Privileges:

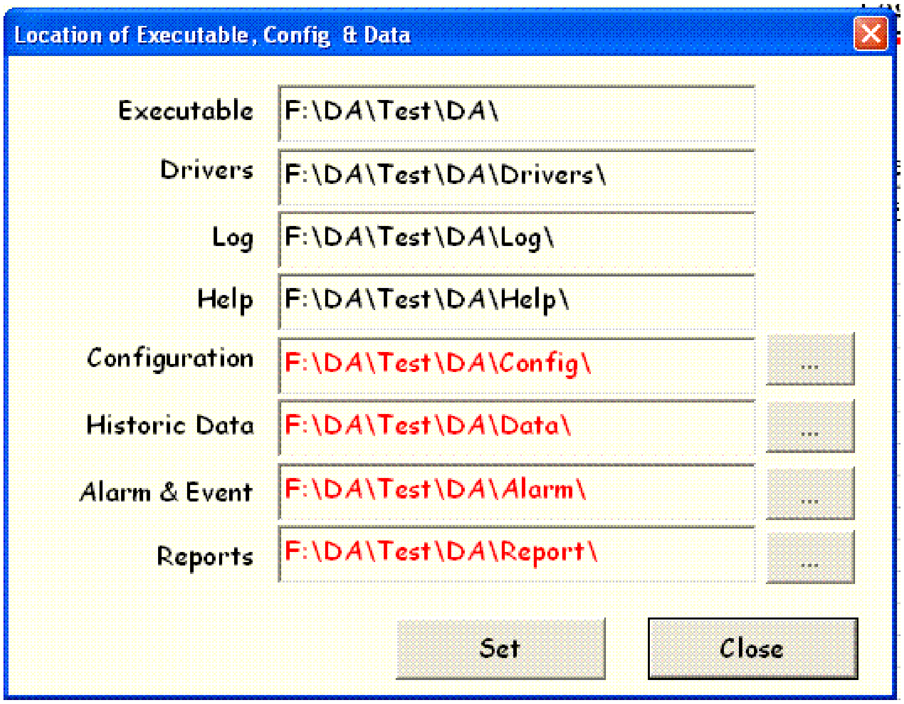
At present users with Administrative privileges have the following rights.

- Only Administrative Privileged users can change the Channel Addresses. So *Set Address* button in the front panel is set active only if the login user has an administrative privilege.
- In the Alarm & Event view, only Administrative privileged users can delete any of the generated alarm. So the *Delete* button in the Alarms& Event view is active if the currently login user have administrative privilege.
- The User Management dialog is accessible only for Administrative privileged users.

7. Other Settings:

7.1 Directory (Location) Configuration:

For storing Historic data, Alarms and Event and Reports there will be separate directories for each data type. You can change these directory locations by selecting the Paths menu item from the Settings menu. It will display the following Locating settings dialog.



In this directory structure you can't change the directories for Executable, Drivers, Log and Help locations. But you can change the location for Configuration, Historic Data, Alarm & Event and Reports location. Once these locations change all the data, alarms and configuration files will be automatically stored in the new directory locations.

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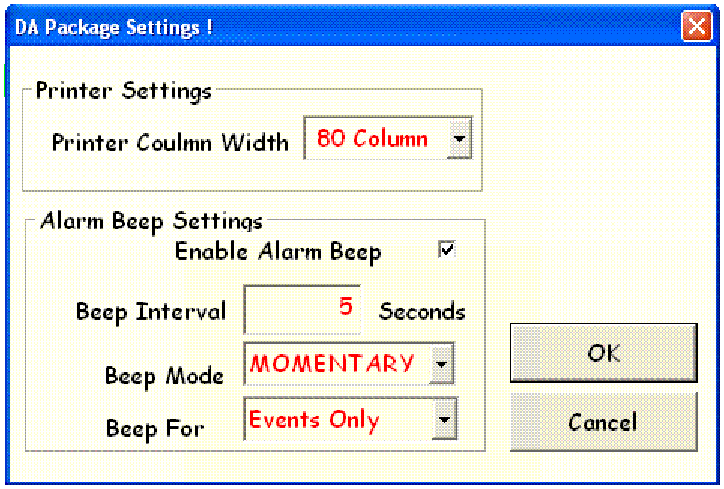
7.2 Printer & Alarm Beep Settings:

To set the Printer and Alarm Beep related setting you can use this "*Printer & Alarm Beep Settings*". Users can display this dialog by simply clicking on *Settings → Printer & Alarm Beep Settings*. This dialog is shown in below fig:



Printer Column Width: At present for Printer this package supporting to set the column width only. User can select the column width of their printers. It can be either 80 column or 132 columns.

Enable Alarm Beep: If user checks this check box, whenever alarms occurred at user defined beep interval the system makes alarm sound.

Beep Interval: Interval to make Alarm sound.

Beep Mode: There will be two types of alarms available. They are *MOMENTARY* & *LATCH*.

- *A **Momentary Alarm*** makes sound, only if the PV is in alarm state. If that channel is recovered from alarm state it can't make any alarm sound.
- A ***Latch Alarm*** makes sound even if the PV is recovered from alarm state until the user acknowledges the alarm.

Ex: if we wish to alarm when the PV exceeds 90. In the case of *Momentary* the alarm will trigger as long as the signal is greater then 90 and will stop sounding when the signal drops below the alarm value (90). But in the case of *Latched* alarm it will makes alarm sound even if the PV drops below the alarm value. It will stops when the operator acknowledges it.

Beep For: You can configure to make alarm sound Only for Alarms, Only for Events or Alarms & Events. No Alarm beep service for Messages.

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8. Communication Protocols

At presently this package is shipping with the following set of protocol drivers:

- OPC Client
- Modbus Ethernet
- Modbus Serial – RTU
- Modbus Serial – ASCII

Users of this package can develop the protocol drivers for their devices have custom protocols that can communicate with various communication mediums (*Like USB, PCI, etc.,*) and they can start accessing the data from the device with this package. To develop custom drivers your package vendor can provide driver development wizard.

For more details on Modbus protocol specifications, visit www.modbus.org
For More details on OPC specifications, visit www.opcfoundation.org

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